



Nonlinear Stability in a Free Boundary Model of Active Locomotion

LEONID BERLYAND, C. ALEX SAFSTEN  & LEV TRUSKINOVSKY

Communicated by M. ORTIZ

Abstract

Contraction-driven self-propulsion of a large class of living cells can be modeled by a Keller-Segel system with free boundaries. The ensuing “active” system, exhibiting both dissipation and anti-dissipation, features stationary and traveling wave solutions. While the former represent static cells, the latter describe propagating pulses (solitary waves) mimicking the autonomous locomotion of the same cells. In this paper we provide the first proof of the asymptotic nonlinear stability of both of these solutions, static and dynamic. In the case of stationary solutions, the linear stability is established using the spectral theorem for compact, self-adjoint operators, and thus linear stability is determined classically, solely by eigenvalues. For traveling waves the picture is more complex because the linearized problem is non-self-adjoint, opening the possibility of a “dark” area in the phase space which is not “visible” in the purely eigenvalue/eigenvector approach. To establish linear stability in this case we employ spectral methods together with the Gearhart-Prüss-Greiner (GPG) theorem, which controls the entire spectrum via bounds on the resolvent operator. For both stationary and small-velocity traveling wave solutions, nonlinear stability is then proved for appropriate parameter values by showing that the nonlinear part of the problem is dominated by the linear part and then employing a Grönwall inequality argument. The developed novel methodology can prove useful also in other problems involving non-self-adjoint (non-Hermitian or non-reciprocal) operators which are ubiquitous in the modeling of “active” matter.

1. Introduction

The ability of cells to self-propel is fundamental for many aspects of development, homeostasis, and disease, for instance, cells need to move to form tissues and their migration is also critical during tissue repair [31, 91, 93, 99]. The active machinery behind self-propulsion resides in the cytoskeleton—a meshwork of actin

filaments with contractile cross-linkers represented by myosin motors. The main active processes in the cytoskeleton are the polymerization of actin fibers and the relative sliding of actin fibers induced by myosin motors [2]. The molecular and biochemical basis of these processes is basically known, however the corresponding mathematical theory is still under development and a variety of multiscale simulation approaches targeting various cell motility mechanisms can be found in the literature [8, 18, 20, 21, 47, 58, 70, 78, 98, 103].

Aiming at the development of a rigorous mathematical approach to stability analysis of such models, we focus in this paper on the simplest phenomenon of self-propulsion in a particular class of cells: keratocytes. They move by advancing the front through polymerization with a simultaneous formation of adhesion clusters. After the adhesion of the protruding part of the cell is secured, the cytoskeleton contracts due to activity of myosin motors. This contraction leads to detachment at the rear and depolymerization of the actin network. All three components of the motility mechanism (polymerization, contraction, and adhesion) depend upon continuous ATP hydrolysis and require intricate regulation by complex signaling pathways involving chemical and mechanical feedback loops [10, 95].

Contractile force generation is of fundamental importance for this mode of cell migration. Using actin fibers as a substrate, myosin motors [51] generate forces which are ultimately responsible for both the motility initiation and the steady locomotion of keratocytes [1, 24, 44, 68, 97]. In view of such central role of active contraction and to achieve relative analytical transparency of the mathematical analysis, we consider in this paper a prototypical model which emphasizes contraction as the main driving mechanism while accounting for polymerization and adhesion only in a schematic manner.

Our minimal model of cell motility is based on a one-dimensional projection of the complex intracellular dynamics onto the direction of motion. More specifically, we assume that the motor part of a cell can be viewed as a one-dimensional continuum with two free boundaries representing the front and the rear of the moving cell. We make a simplifying physical assumption that actin polymerization and depolymerization can take place only on these boundaries and that these phenomena can be modeled as an influx of mass at the front boundary and its disappearance at the rear boundary. The adhesion is also treated in an over-simplified form as passive spatially inhomogeneous viscous friction. Instead, the actomyosin contraction, which is the main player, is represented by active spatially inhomogeneous prestress [53, 61].

As it was first shown in [79, 80], the mathematical model, which captures all these physical effects while being amenable to rigorous mathematical analysis, reduces to the one-dimensional Keller-Segel system with free boundaries. In contrast to the conventional chemotactic Keller-Segel model [56], here the same set of equations emerges in a purely mechanical setting; see [16, 19, 33, 53, 61, 63, 82] for the earlier insights along the same lines. In Sect. 2, where we present for convenience a short derivation of this model, we also highlight its universality (minimality) by showing that it can be obtained starting from rather different physical assumptions.

It is important to mention that alternative free-boundary-type models of cell motility, emphasizing various other components of the self-propulsion machinery,

have been used in numerical simulations [57, 71, 72, 83, 89] and, in some cases, also subjected to rigorous mathematical analysis [28, 29]. Closely related free boundary models describing tumor growth have been also studied both analytically and numerically [30, 45, 46, 49]. Our paper differs from all this mathematical work on free boundary modeling of locomotion in its emphasis on the non-self-adjoint property of the linearized operator resulting from both nonlocality [52] and activity [5]. Note that phase field models of cell motility, representing a mathematical proxy to our free boundary formulation (front capturing instead of front tracking [15, 32]), have been also a subject of extensive research efforts [12–14, 102, 104]. However, while the corresponding models allow for very efficient numerical simulations, they are usually not as readily amenable for rigorous stability analysis, and therefore will not be addressed in the present purely analytical study.

The one-dimensional Keller-Segel system with free boundaries is known to possess a family of pulse-like traveling wave solutions, which describe steady autonomous locomotion of individual cells [79, 80, 86, 87]. These solutions, which can be interpreted as solitary waves, bifurcate from a family of stationary (static) solutions, representing non-moving cells. The role of bifurcation parameter is played by a non-dimensional measure of the level of internal activity with both static and dynamic solutions being “active” in the sense that they consume and dissipate energy. In [79, 80] the whole variety of stationary solutions was constructed analytically and the nature of the corresponding static-dynamic bifurcation was determined using weakly nonlinear analysis involving a standard approach based on Lyapunov-Schmidt reduction [48]; significant numerical evidence that traveling waves bifurcating from homogeneous stationary states have finite reserve of stability was also obtained. In [86, 87], the same bifurcation between stationary and traveling wave solutions was studied in two dimensions, and the configurations of the traveling wave solutions were computed both analytically (close to the bifurcation point) and numerically (away from it). Linear stability was addressed for both stationary and traveling wave solutions with the eigenvalue-based stability condition computed explicitly.

1.1. Summary of the Main Results

The present paper begins with a derivation of the free-boundary model of cell motion in Sect. 2 followed by nondimensionalization and a further derivation of the “stiff limit” of this model as the cell’s elastic stiffness (see (3)) becomes infinite in Sect. 3. The ultimate model on which the bulk of our analysis is focused is

$$\begin{cases} \partial_t m = m'' + \phi'(1/2)m' - (m\phi')' & -1/2 < x < 1/2 \\ -Z\phi'' + \phi = Pm & -1/2 < x < 1/2, \end{cases} \quad (1)$$

where m represents myosin density and satisfies Neumann boundary conditions and ϕ represents pressure and satisfies periodic boundary conditions (here, the prime $'$ denotes spatial derivative). The remaining parameters in the model are the magnitude of viscosity force (viscosity of the cytogel) in the cell denoted by Z and the magnitude of activity force (activity rate of the myosin) in the cell, denoted P .

In Sect. 4, we focus on homogeneous stationary solutions to the model. The main achievement of this section is Theorem 4.1, which can be summarized as follows:

Nonlinear stability of stationary solutions: The stationary solutions of (1) are nonlinearly exponentially stable when the physical parameters P, Z are in the range that is explicitly determined by a transcendental equation.

In Sect. 5, we examine traveling wave solutions to the model, showing in Theorem 5.5 that a family of traveling waves bifurcates from the stationary solution at $P = P_0$ via the Crandall-Rabinowitz (CR) theorem [27]. We describe this family of traveling wave solutions parameterized by their velocity V and the corresponding activity parameter $P_{TW}(V)$ which characterizes the total amount of activity of myosin motors required to move with velocity V .

Finally, Sect. 6 is focused on stability of traveling wave solutions, with the primary result of this paper being Theorem 6.1, can be which is summarized as follows:

Nonlinear stability of traveling waves via non-self-adjoint spectral analysis:

There exists $Z^*, V^* > 0$ so that if $Z > Z^*, |V| < V^*$ and $P = P_{TW}(V)$, then the traveling wave with velocity V is exponentially stable.

The key observation in the proof of this theorem is that the standard stability analysis based on eigenvalues and eigenvectors is not sufficient. Indeed, due to non-self-adjointness of the linearized operator, eigenvectors may not span the entire phase space and alternative techniques based on resolvent analysis were developed. The second challenge in this proof is the transition from linear to nonlinear stability. Evaluated at an arbitrary perturbation of the traveling wave solution, the nonlinear operator for this model can be written as its linearization about the traveling wave plus a nonlinear part bounded by the product of the L^2 and H^1 norms of the perturbation. We then show via a series of subtle bounds (reminiscent of parabolic regularization) that the H^1 norm of a solution to (1) can be controlled by the L^2 norm provided the H^1 norm is small at $t = 0$. This allows us to show that the linear part of (1) dominates the nonlinear part in the vicinity of the traveling wave, allowing the linear stability to be used to prove nonlinear stability.

1.2. Methods and Challenges

The main difficulty in the stability analysis of the traveling wave solutions resides in the non-self-adjoint (non-Hermitian, or non-reciprocal) nature of the corresponding linearized operator [35], which is an important general feature of PDE models of “active” matter [5, 34, 39, 92, 101]. It is known, for instance, that for non-self-adjoint (NSA) operators, eigenvectors do not necessarily span the entire domain of the operator. Therefore, common stability analysis, e.g. [3, 73], based only on eigenvalues and eigenvectors may not be sufficient [96].

We recall that when the linearized problem is self-adjoint, the eigenmodes of the stable system can be divided into stable (corresponding to eigenvalues with negative real part), and center (with zero real part eigenvalues). In the nonlinear setting, solutions in the corresponding stable manifold would then be controlled (bounded) by solutions in the center manifold. Furthermore, a nonlinear ODE can be

derived for solutions in the center manifold, from which it can be shown that all such solutions asymptotically approach the equilibrium. It would then mean that all other solutions also approach it. The key assumption in this approach to stability is that eigenvectors of the linearized operator span the entire domain of the operator. This may not be the case for NSA operators which typically exhibit a “dark” area in the phase space which is not “visible” in the purely eigenvalue/eigenvector approach. We address this challenge using directly resolvent analysis instead of relying solely on eigenvalues.¹

In the NSA case, where we have to deal with the entire spectrum of the linearized operator, linear stability can be established by applying the Gearhart-Prüss-Greiner (GPG) theorem [41] which operates directly with bounds on the resolvent of the linear operator. Specifically, when eigenvectors do not span the domain of the operator A , the GPG theorem turns to the analysis of another operator

$$R_\mu = (\mu I - A)^{-1} \quad (2)$$

with the parameter μ having a positive real part. The crucial step is then to bound R_μ away from any point of the entire spectrum, not just the eigenvalues. In particular, even in infinite dimensions, such a bound rules out the cases when a sequence of eigenvalues has negative real parts converging to zero.

After the linear stability is established, a natural step in checking the nonlinear stability would be, at least in finite dimensions, to use the Hartman-Grobman (HG) theorem [4]. However, even in this case, this theorem requires the absence of eigenvalues with zero real part. Our problem has a zero eigenvalue (a slow manifold) which appears in the linearized operator due to translational symmetry. To overcome this complication, we use the notion of “stability up to shifts”, see for instance [87], and prove the appropriate analog of the HG theorem specifically tailored for our infinite dimensional problem. While there are several extensions of the HG theorem to infinite dimensions, e.g. [6], most of these results apply to a smooth nonlinear operator mapping a Banach space to itself whereas in our parabolic PDE problem, the operator maps a Sobolev space H^2 to L^2 . The existing HG type results for parabolic equations [67] are also not directly applicable to our problem. Our original approach is based on establishing subtle bounds on the derivatives of the solution in the neighborhood of a pitchfork bifurcation which allow one to decide when the linear part of the nonlinear operator dominates its nonlinear part. Our result is then equivalent to establishing the existence of a Lyapunov function (or rather Lyapunov functional in our infinite dimensional setting) for the pulse-like traveling wave solutions with synchronously moving free boundaries, see [11, 69, 90] for related results. We emphasize that our approach is readily generalizable to other PDE models where the task is to prove exponential stability of an emerging nontrivial solution in the vicinity of a bifurcation point.

While our approach is original, it is important to mention that a large variety of other methods for establishing nonlinear stability of traveling waves have been

¹ It should be noted that in some specific NSA problems, eigenvectors do span the whole domain and in those cases the absence of negative eigenvalues may be sufficient for stability, e.g. [62].

explored in the literature, see the reviews in [55, 75, 88]. In particular, several studies deal specifically with spectral stability of traveling wave solutions by showing that the spectrum of the linearized operator consists only of points with negative real part [64, 65, 76]. Most of these studies use the method of Evans function, which is a convenient tool for separating the eigenvalues from the continuous spectrum ubiquitous in traveling wave problems defined in unbounded domains [7, 23, 42]. We do not use the Evans function based approach for two reasons. First, our traveling waves are compact and there is no continuous spectrum for our problem. Second, in our specific problem, we can circumvent the use of Evans function by resorting to a simpler approach to calculate the leading eigenvalue developed in [26]. Other studies of linear and nonlinear stability of traveling waves, which use spectral theory to obtain bounds on the semigroup generated by the linear operator and then showing that the nonlinear problem is dominated by the linearization, can be found in [23, 54, 59]. While we basically follow the same strategy, our main spectral theoretic tool, which is the GPG theorem, is different from all those used in the previous studies.

2. The Model

In this Section we briefly explain how our one-dimensional Keller-Segel system with free boundaries can be derived from physical considerations. To emphasize that this model is both minimal and universal, we present two alternative derivations based on apparently contradicting assumptions that the material inside the cell is either infinitely compressible or infinitely incompressible.

In the original, infinitely compressible version of the model, proposed in [77, 79, 80], we start by writing the 1D force balance for a gel segment in the form

$$\sigma' = \xi v,$$

where $\sigma(x, t)$ is axial stress, $v(x, t)$ is the velocity of the gel, ξ is the coefficient of viscous friction. We denote a single spatial coordinate by x and time by t ; prime denotes the spatial derivative. The assumption of infinite compressibility of the gel decouples the force balance equation from the mass balance equation. Specifically, by neglecting compressibility, we can write the constitutive relation for an active gel, representing the material inside the cell, in the form

$$\sigma = \eta v' + km,$$

where η is the bulk viscosity, $m(x, t)$ is the mass density of myosin motors and $k > 0$ is a constant representing contractile pre-stress per unit motor mass. The density of motors is modeled by a standard advection–diffusion equation where the advection is perceived to be originating from the flow of actin [16, 50, 100] i.e.,

$$\partial_t m + (mv)' - Dm'' = 0,$$

where D is the constant effective diffusion coefficient, see [80] for the discussion of its physical meaning. Behind this equation is the assumption that myosin motors,

actively cross-linking the implied actin meshwork, are not only being advected by the network flow but can also diffuse due to the presence of thermal fluctuations. To ensure that the moving cell maintains its size, we follow [9,38,66,77,81] and introduce a phenomenological cortex/osmolarity mediated quasi-elastic non-local interaction linking the front and the back of the self-propelling fragment. More specifically, we assume that the boundaries of our moving active segment interact through an effective linear spring which regulates the value of the stress on the free boundaries $l_-(t)$ and $l_+(t)$:

$$\sigma_0^\pm = -k_e(L(t) - L_0)/L_0. \quad (3)$$

Here $L(t) = l_+(t) - l_-(t)$ is the length of the moving segment, k_e is the effective elastic stiffness and L_0 is the reference length of the spring. Finally, we assume that the free boundaries are not penetrable which means that they move with the internal flow. Therefore the following kinematic boundary condition must hold: $\partial_t l_\pm = v(l_\pm)$. Our assumptions allow us to avoid addressing explicitly the mass balance equation since we effectively postulate that the addition of actin particles at the front is fully compensated by their synchronous removal at the rear, see [79,80] for details. We also impose a zero flux condition for the active component $m'(l_\pm(t), t) = 0$ ensuring that the average concentration of motors $m_0 = L_0^{-1} \int_{l_-(t)}^{l_+(t)} m(x, t) dx$ is preserved. To complete the setting of the ensuing mechanical problem we impose the initial conditions $l_\pm(0) = l_\pm^0$ and $m(x, 0) = m^0(x)$. One can see that the resulting one-dimensional problem is equivalent to a dynamical system of a Keller–Segel-type with free boundaries, however, in contrast to the chemotactic analog, the nonlocality here is mechanical rather than chemical in origin.

Next, we derive the same model under the incompressibility assumption as it was first proposed in [86,87]. This alternative derivation emphasizes the paradigmatic nature of our problem. We also switch now to a more formal mathematical description as appropriate for the subsequent rigorous analysis.

The cell, whose cytoskeleton is now viewed as an incompressible fluid, is again modeled in a time-dependent interval centered at a point $c(t)$ and with width $L(t)$: $\Omega(t) = (c(t) - L(t)/2, c(t) + L(t)/2) \subset \mathbb{R}$. For each $t \geq 0$, the velocity of fluid in the cell is $u(x, t) \in \mathbb{R}$ for $x \in \Omega(t)$. Since the cell is thin in the dorsal direction, most of the fluid within the cell lies close to the cell membrane. Therefore, we may assume the flow of incompressible fluid is dominated by friction and follows Darcy's law:

$$p' = \zeta u.$$

Here, $p(\cdot, t) : \Omega(t) \rightarrow \mathbb{R}$ is the total pressure within the cell, $\zeta > 0$ is the constant adhesion coefficient and the prime denotes spatial derivative; see Appendix D in [22] for the detailed physical derivation. Following [87], we write the equation for the total pressure in the form

$$p = \mu u' + km,$$

where $m(\cdot, t) : \Omega(t) \rightarrow \mathbb{R}_+$ is the density of myosin within the cell, $\mu > 0$ is the viscosity coefficient, and $k > 0$ is the contractility of myosin. The incompressibility assumption suggests that the hydrostatic fluid pressure here should be viewed as a kinematic variable which in our 1D setting is constant and can be simply absorbed into p .

We further assume that, again, myosin density evolves in time according to the advection–diffusion equation, which, after applying Darcy’s law can be written in the form (see [87] for details):

$$\partial_t m = Dm'' - (m\phi')'. \quad (4)$$

To ensure that the total myosin mass is conserved in time, the myosin density satisfies no-flux boundary conditions: $m' = 0$.

A boundary condition for total pressure is obtained from an assumption that there is a non-constitutive global elastic restoring force due to osmotic effects or the cell membrane cortex tension. This non-local effect is modeled by the condition

$$p = -k_e(L(t) - L_0)/L_0, \quad (5)$$

where L_0 is the length of the cell in a reference configuration, and $k_e > 0$ is the elastic stiffness coefficient. Note that in view of the possibility for the fluid to escape in the dorsal direction, our 1D cell is effectively compressible despite the incompressibility of the fluid, so that the coefficient k_e can be also interpreted as the inverse compressibility of the cell as a whole.

The ensuing 1D problem is analytically transparent due to the fact that the variable $\phi = p/\zeta$ satisfies a linear elliptic equation:

$$-\mu\phi'' + \zeta\phi = km. \quad (6)$$

The complexity of the problem resides in the boundary conditions. Thus, as we have seen, the variable ϕ at the boundary satisfies the nonlocal condition of Dirichlet type. Furthermore, to find the time evolution of the interval $\Omega(t)$, we assume that both components of our binary mixture, solvent (actin) and solute (myosin), respect the kinematic (velocity-matching) boundary conditions of Hele-Shaw type:

$$\partial_t L(t) = \phi'(c(t) + L(t)/2, t) - \phi'(c(t) - L(t)/2, t) \quad (7)$$

$$\partial_t c(t) = \frac{1}{2} (\phi'(c(t) + L(t)/2, t) + \phi'(c(t) - L(t)/2, t)). \quad (8)$$

While the resulting fluid model, introduced in [86,87], is mathematically equivalent of the solid gel model introduced in [79,80], it is the Hele-Shaw fluid formulation which opened the way towards two-dimensional analysis allowing one to track not only the position of a moving cell but also the evolution of its shape; the corresponding 2D version of the gel model can be found in [43].

3. Infinite Rigidity Limit

Below we distinguish between two versions of our model of cell locomotion. The first one, to which we refer as “Model A”, assumes, as in the original derivation, that the size of the cell is controlled by an elastic spring, whose elastic modulus will be referred to as the “stiffness” parameter. By considering an asymptotic limit when such stiffness tends to infinity, we will derive a “stiff limit” of the original model to which we refer in what follows as “Model B.” In this limiting model, which turns out to be analytically much more transparent, the cell has a fixed size.

We recall that in the model A, the myosin density m , pressure ϕ , length L , and center c satisfy the PDE system:

$$\left\{ \begin{array}{ll} -\mu\phi'' + \zeta\phi = km & x \in \Omega(t) \end{array} \right. \quad (9)$$

$$\left\{ \begin{array}{ll} m_t = Dm'' - (m\phi')' & x \in \Omega(t) \end{array} \right. \quad (10)$$

$$\left\{ \begin{array}{ll} \zeta\phi = -k_e \frac{L(t) - L_0}{L_0} & x \in \partial\Omega(t) \end{array} \right. \quad (11)$$

$$\left\{ \begin{array}{ll} m' = 0 & x \in \partial\Omega(t) \end{array} \right. \quad (12)$$

$$\partial_t L(t) = \phi'(c(t) + L(t)/2, t) - \phi'(c(t) - L(t)/2, t) \quad (13)$$

$$\left\{ \begin{array}{ll} \partial_t c(t) = \frac{1}{2} (\phi'(c(t) + L(t)/2, t) + \phi'(c(t) - L(t)/2, t)) \end{array} \right. \quad (14)$$

An important feature of this model is that total myosin mass is conserved in time:

$$\frac{d}{dt} \int_{c(t)-L(t)/2}^{c(t)+L(t)/2} m(x, t) dx = 0. \quad (15)$$

To nondimensionalize this model, we rescale x by $x \rightarrow x/L_0$ and accordingly $L \rightarrow L/L_0$, $c \rightarrow c/L_0$. We then rescale time by $t \rightarrow tD/L_0^2$, pressure by $\phi \rightarrow \phi\zeta/k_e$ and myosin density by $m \rightarrow mL_0^2/M$ where M is the total myosin mass.

After such normalization, the variables x, t, m, ϕ, L , and c are all dimensionless quantities and the PDE system (9)–(14) can be re-written in the form

$$\left\{ \begin{array}{ll} -Z\phi'' + \phi = Pm & c - L/2 < x < c + L/2 \end{array} \right. \quad (16)$$

$$\left\{ \begin{array}{ll} m_t = m'' - K(m\phi')' & c - L/2 < x < c + L/2 \end{array} \right. \quad (17)$$

$$\left\{ \begin{array}{ll} \phi = 1 - L & x = c \pm L/2 \end{array} \right. \quad (18)$$

$$\left\{ \begin{array}{ll} m' = 0 & x = c \pm L/2 \end{array} \right. \quad (19)$$

$$\left\{ \begin{array}{ll} L_t = K(\phi'(c + L/2) - \phi'(c - L/2)) & \end{array} \right. \quad (20)$$

$$\left\{ \begin{array}{ll} c_t = K \frac{\phi'(c + L/2) + \phi'(c - L/2)}{2} \end{array} \right. \quad (21)$$

Here we introduced the main non-dimensional parameters of the model

$$Z = \frac{\mu}{\zeta L_0^2} \quad P = \frac{kM}{k_e L_0} \quad K = \frac{k_e}{D\zeta}, \quad (22)$$

representing dimensionless measures of bulk viscosity, activity level and the non-local stiffness, respectively. Note that in these rescaled coordinates, the total myosin mass is

$$\int_{c(t)-L(t)/2}^{c(t)+L(t)/2} m(x, t) \, dx = 1. \quad (23)$$

A simpler and analytically more tractable model can be obtained if we consider the limit when the “stiffness” coefficient k_e tends to infinity, see also [80]. To this end, suppose $k_e = k_e^*/\varepsilon$ where ε is a small, positive constant. Each of the coefficients P, K in (22) depend on k_e . Therefore, we denote them $P_\varepsilon = \varepsilon P_1$, and $K_\varepsilon = K_{-1}/\varepsilon$ respectively where

$$P_1 = \frac{kM}{k_e^* L_0} \quad K_{-1} = \frac{k_e^*}{D\zeta}. \quad (24)$$

For each $\varepsilon > 0$, we now assume that $\phi_\varepsilon, m_\varepsilon, L_\varepsilon$, and c_ε solve (16)–(21) with coefficients

P_ε , and K_ε for $0 \leq t < T$ with initial conditions $m_\varepsilon(0, x) = \bar{m}(x)$, $L_\varepsilon(0) = \bar{L}$, and $c_\varepsilon(0) = \bar{c}$. We then expand each of $\phi_\varepsilon, m_\varepsilon, L_\varepsilon$, and c_ε in small ε :

$$\phi_\varepsilon = \phi_0 + \varepsilon \phi_1 + O(\varepsilon^2) \quad m_\varepsilon = m_0 + \varepsilon m_1 + O(\varepsilon^2) \quad (25)$$

$$L_\varepsilon = L_0 + \varepsilon L_1 + O(\varepsilon^2) \quad c_\varepsilon = c_0 + \varepsilon c_1 + O(\varepsilon^2). \quad (26)$$

The next step is to substitute these expansions along with the expansions for P_ε and K_ε into our equations and compare terms of like power in ε . Note that the free boundary requires that the boundary conditions be expanded not only in ε , but also x so that all boundary conditions are evaluated at $x = c_0 \pm L_0/2$.

In zeroth order, (16) becomes $-Z\phi_{0,xx} + \phi_0 = 0$ with boundary condition $\phi_0 = 1 - L_0$. We explicitly calculate that

$$\phi_0 = (1 - L_0) \frac{\cosh\left(\frac{c_0 - x}{\sqrt{Z}}\right)}{\cosh\left(\frac{L_0}{2\sqrt{Z}}\right)}. \quad (27)$$

In order -1 , (17) has only one nontrivial term:

$$K_{-1}(m_0\phi_{0,x})' = 0. \quad (28)$$

We conclude that $m_0\phi_{0,x}$ is constant in x . Since the initial condition $\bar{m}$ is arbitrary in $H^2(-L_0/2, L_0/2)$, and since ϕ_0 given by (27) does not depend on m_0 , this can only be accomplished if $\phi_{0,x} = 0$. This, in turn, implies $\phi_0 = 0$ and $L_0 = 1$.

Since $\phi_0 = 0$, in zeroth order, (17) becomes

$$m_{0,t} = m_{0,xx} - K_{-1}(m_0\phi_{1,x})'. \quad (29)$$

Here ϕ_1 satisfies the first order expansions of (16) and its boundary condition (18):

$$\begin{cases} -Z\phi_{1,xx} + \phi_1 = P_1 m_0 & c_0 - L_0/2 < x < c_0 + L_0/2 \\ \phi_1 = -L_1 & x = c_0 \pm L_0/2 \end{cases} \quad (30)$$

In zeroth order, (20) becomes

$$L_{0,t} = K_{-1}(\phi_{1,x}(c_0 + L_0/2) - \phi_{1,x}(c_0 - L_0/2)). \quad (31)$$

On the other hand, we know that $L_0 \equiv 1$, so $L_{0,t} = 0$. Therefore, ϕ_1 has to satisfy three boundary conditions:

$$\begin{aligned} \phi_1(c_0 - L_0/2) &= -L_1, \quad \phi_1(c_0 + L_0/2) = -L_1, \\ \phi_{1,x}(c_0 - L_0/2) &= \phi_{1,x}(c_0 + L_0/2). \end{aligned} \quad (32)$$

We conclude that L_1 is determined by these three conditions. Since ϕ_1 satisfies periodic boundary conditions, we may determine c_0 by the zeroth order expansion of (21):

$$c_{0,t} = K_{-1}\phi_{1,x}(c_0 + L_0/2). \quad (33)$$

To complete the derivation of our “Model B”, we omit the subscript indices so we write $m = m_0$, $L = L_0 = 1$, $c = c_0$. It is also convenient to denote $\phi = K_{-1}\phi_1$ and introduce a new dimensionless parameter

$$P = P_1 K_{-1}.$$

We can then formulate the stiff limit of our model in the form of a reduced PDE system:

$$\begin{cases} -Z\phi'' + \phi = Pm & c(t) - 1/2 < x < c(t) + 1/2 \end{cases} \quad (34)$$

$$\begin{cases} m_t = m'' - (m\phi')' & c(t) - 1/2 < x < c(t) + 1/2 \end{cases} \quad (35)$$

$$\begin{cases} m' = 0 & x = c(t) \pm 1/2 \end{cases} \quad (36)$$

$$\begin{cases} \phi(c - 1/2) = \phi(c + 1/2) \end{cases} \quad (37)$$

$$\begin{cases} \phi'(c - 1/2) = \phi'(c + 1/2) \end{cases} \quad (38)$$

$$\begin{cases} c_t = \phi'(c + 1/2). \end{cases} \quad (39)$$

Once again, the total myosin mass is conserved and its value is 1. Due to its relative simplicity, combined with the ability to still represent the main physical effects, model (34)-(39), first introduced formally in [80], will be the main focus of the present study.

Note first that Model B has stationary solution

$$m = 1, \quad \phi = P.$$

As we are going to see, for P sufficiently small, such solution is exponentially stable, but for large P , it becomes unstable and bifurcates into traveling wave solutions. In the linearization of Model B about traveling waves, the invariance of traveling waves to translation manifests itself through a zero eigenvalue. By factorizing the

shifts, the corresponding eigenvector is identified with 0, and in this way the zero eigenvalue is effectively removed.

The factorization of the shifts is accomplished by changing coordinates via the transformation $x \rightarrow x - c(t)$. In the new coordinates, (34)–(39) become

$$\begin{cases} -Z\phi'' + \phi = Pm & -1/2 < x < 1/2 \end{cases} \quad (40)$$

$$\begin{cases} m_t = m'' + \phi'(1/2)m' - (m\phi')' & -1/2 < x < 1/2 \end{cases} \quad (41)$$

$$\begin{cases} m' = 0 & x = \pm 1/2 \end{cases} \quad (42)$$

$$\begin{cases} \phi(-1/2) = \phi(1/2) \end{cases} \quad (43)$$

$$\begin{cases} \phi'(-1/2) = \phi'(1/2) \end{cases} \quad (44)$$

$$\begin{cases} c_t = \phi'(1/2). \end{cases} \quad (45)$$

Observe that the center coordinate c is partially decoupled from m . Therefore, we can drop the c coordinate, effectively grouping together all solutions that are the same up to a translation of the center in particular, a stationary solution to (40)–(44) becomes an element of an equivalence class of traveling wave solutions to (34)–(39). Furthermore, we see that if a stationary solution to (40)–(45) is exponentially stable, then the corresponding traveling wave solutions to (34)–(39) are also exponentially stable “up to shifts”. Here we imply that a solution whose initial condition is a perturbation of a traveling wave solution may converge to a different traveling wave solution with the same velocity, but a different (“shifted”) center coordinate.

In what follows we refer to the result of transforming coordinates in Model B and dropping the c component as “Model C,” which we formulate more succinctly

$$\begin{aligned} \partial_t m &= F_C(m) = m'' + \phi'(1/2)m' - (m\phi')', \\ \begin{cases} -Z\phi'' + \phi = Pm & -1/2 < x < 1/2 \\ \phi(1/2) = \phi(-1/2) \\ \phi'(-1/2) = \phi'(1/2). \end{cases} \end{aligned} \quad (46)$$

The domain of the nonlinear operator F_C is

$$X_C^2 := \left\{ m \in H^2(-1/2, 1/2) : m'(\pm 1/2) = 0, \int_{-1/2}^{1/2} m \, dx = 1 \right\}.$$

4. Nonlinear Stability of Stationary Solutions

We can now address the stability (up to shifts) of homogeneous stationary solutions $m_S := 1$ and $\phi = P$ of model B, by analyzing the $m_S = 1$ solution to model C its; nonlinear exponential stability is captured in the following theorem:

Theorem 4.1. *Let $Z > 0$. Then there exist $P_0 > \pi^2 Z$ and $\varepsilon, r > 0$ such that if $0 < P < P_0$ and $m_0 \in H^1(-1/2, 1/2)$ with $\|m_0 - m_S\|_{H^1} < \varepsilon$, then for any solution $m(x, t)$ to $\partial_t m = F_C(m)$ with $m(0, x) = m_0(x)$ we have*

$$\|m(\cdot, t) - m_S\|_{L^2} \leq \|m_0 - m_S\|_{L^2} e^{-rt}. \quad (47)$$

Remark 4.2. In Sect. 5, we will find $P_0 = P_0(Z)$ as a solution to the explicit transcendental equation (96). Thus, Theorem 4.1 can be stated as a guarantee of exponential stability of m_S provided the model parameters P and Z are in the set $\{(P, Z) : 0 < Z \text{ and } 0 < P < P_0(Z)\}$ where P_0 can be computed numerically.

A physical interpretation of Theorem 4.1 is that if the activity rate P is low enough relative to the viscosity Z , then the cell cannot move, and so stationary solutions in our model are stable. The proof of the nonlinear stability Theorem 4.1 can be broken down into two parts: (1) a proof of linear stability, and (2) a proof that close to the stationary solution, the nonlinear part of the model is dominated by the linear part. The former is largely a classical proof, though we include all details for completeness. The latter is a novel proof centered on the Gagliardo-Nirenberg theorem and the concept of diffusive regularization.

We begin the process of the proof of Theorem 4.1 with a study of linear stability. The linearization of model C is

$$S_C u = D F_C(1)u = u'' - \phi'', \quad (48)$$

where ϕ is defined as in (46) (with $m = u$) and $u \in \tilde{X}_C^2 := \{u \in H^2(-1/2, 1/2) : u'(\pm 1/2) = 0, \int_{-1/2}^{1/2} u \, dx = 0\}$. We first prove the following theorem establishing the linear stability of stationary states:

Theorem 4.3. *Let $Z > 0$. Then there exist $P_0 > \pi^2 Z$ and $\omega > 0$ such that if $P < P_0$ and if $m(x, t)$ solves $\partial_t u = S_C u$ in $\tilde{X}_2$ with $u(0, x) = u_0(x)$, then*

$$\|u(\cdot, t)\|_{L^2} < \|u_0\|_{L^2} e^{-\omega t}. \quad (49)$$

The proof of Theorem 4.3 is classical, and relies on the spectral theorem for compact self-adjoint operators [60], which states that a compact, self-adjoint operator has a basis of eigenvectors. The operator S_C is not compact, but we will show that its inverse is, and S_C shares the eigenvectors of its inverse. Therefore, we may reduce the problem of linear stability to the problem of stability of individual eigenstates, with the exponential decay in Theorem 4.3 given by a uniform negativity of the corresponding eigenvalues. To complete this proof, we must only show (via three Lemmas below) that (i) S_C is self-adjoint, (ii) all eigenvalues of S_C are negative and bounded away from zero, and (iii) S_C has compact inverse. While the proofs of Theorem 4.3 and the supporting results Proposition 4.6 and Lemma 4.7 are classical, we include them for completeness.

First we prove self-adjointness. Let X be a Hilbert space and let $D(A) \subset X$ be a dense subspace of X which is the domain of an operator $A : D(A) \rightarrow X$. Recall that the adjoint of A is an operator A^* such that $\langle u, A^* v \rangle = \langle Au, v \rangle$ for all $u, v \in D(A)$. The operator A is self-adjoint if $A = A^*$. Therefore, we introduce the following bilinear form to determine whether or not an operator is self-adjoint:

Definition 4.4. Let X be an inner product space, and let $A : X \rightarrow X$. The *adjoint commutator* of A is $H : X \times X \rightarrow \mathbb{R}$ defined by

$$H(u, v) = \langle Au, v \rangle - \langle u, Av \rangle. \quad (50)$$

If the adjoint commutator is identically zero, then A is self-adjoint. Otherwise, A is non-self-adjoint.

Lemma 4.5. *The linearization S_C of F_C about the stationary solution $m_S = 1$ is self-adjoint with respect to the L^2 inner product.*

Proof. Let $H : \tilde{X}_2 \times \tilde{X}_2 \rightarrow \mathbb{R}$ be the adjoint commutator of S_C . Let $u_1, u_2 \in \tilde{X}_2$. Let ϕ_i solve $-Z\phi_i'' + \phi_i = Pu_i$ with periodic boundary conditions. Then

$$H(u_1, u_2) = \int_{-1/2}^{1/2} u_2(u_1'' - \phi_1'') dx - \int_{-1/2}^{1/2} u_1(u_2'' - \phi_2'') dx \quad (51)$$

$$= \int_{-1/2}^{1/2} (u_1' u_2' - u_2' u_1') dx + \int_{-1/2}^{1/2} \left[u_1 \frac{\phi_2 - Pu_2}{Z} - u_2 \frac{\phi_1 - Pu_1}{Z} \right] dx \quad (52)$$

$$= \frac{1}{PZ} \int_{-1/2}^{1/2} [(-Z\phi_1'' + \phi_1)\phi_2 - (-Z\phi_2'' + \phi_2)\phi_1] dx \quad (53)$$

$$= \frac{1}{P} \int_{-1/2}^{1/2} (\phi_1' \phi_2' - \phi_2' \phi_1') dx \quad (54)$$

$$= 0. \quad (55)$$

□

Now we show the negativity of the eigenvalues of S_C .

Proposition 4.6. *If $P/Z < \pi^2$, then all eigenvalues of S_C are negative and bounded away from 0.*

Proof. Assume $P/Z < \pi^2$. Let u be an eigenvector of S_C and let λ be its eigenvalue. Since S_C is self adjoint, λ is real and u is real-valued. Without loss of generality, we may assume $\|u\|_{L^2} = 1$. Let ϕ solve $-Z\phi'' + \phi = Pu$ with periodic boundary conditions on $(-1/2, 1/2)$. Then u and λ satisfy $u'' - \phi'' = \lambda u$. Multiplying by u and integrating we find that

$$\begin{aligned} \lambda &= \int_{-1/2}^{1/2} u''u - \phi''u \\ &= - \int_{-1/2}^{1/2} u'^2 dx - \frac{1}{Z} \int_{-1/2}^{1/2} (\phi - Pu)u dx \\ &= - \int_{-1/2}^{1/2} u'^2 dx + \frac{P}{Z} \int_{-1/2}^{1/2} u^2 dx - \frac{1}{PZ} \int_{-1/2}^{1/2} \phi(-Z\phi'' + \phi) dx \\ &= - \int_{-1/2}^{1/2} u'^2 dx + \frac{P}{Z} - \frac{1}{P} \int_{-1/2}^{1/2} \phi'^2 dx - \frac{1}{PZ} \int_{-1/2}^{1/2} \phi^2 dx \\ &\leq - \int_{-1/2}^{1/2} u'^2 dx + \frac{P}{Z}. \end{aligned}$$

It is well known that the optimal constant in the Poincaré inequality in an interval of length 1 is $1/\pi$ [74]. The Poincaré inequality applies to u since $\int_{-1/2}^{1/2} u(x) dx = 0$. Therefore, we conclude that

$$\lambda \leq \frac{P}{Z} - \pi^2 < 0. \quad (56)$$

Therefore all eigenvalues of S_C are negative and bounded away from 0. $\square$

While Proposition 4.6 gives the sufficient condition $P/Z < \pi^2$ for the negativity of the eigenvalues of S_C , it is not optimal in the sense that the eigenvalues of S_C may be negative for P/Z larger than π^2 . However, for fixed Z and sufficiently large P , there are positive eigenvalues of S_C . To see this, observe that $u(x) = \cos(2\pi x)$ is an eigenvector of S_C (since it is also an eigenvector of $-Z\phi'' + \phi$ with periodic boundary conditions on $(-1/2, 1/2)$). Its eigenvalue is $-4\pi^2 + \frac{P}{Z} \left(1 + \frac{1}{4\pi^2 Z}\right)^{-1}$ (note however that this is not the largest eigenvalue). If, for fixed $Z > 0$, the parameter P is large enough, this eigenvalue is positive. This observation hints that for some critical value $P_0 > \pi^2 Z$ of P , the largest eigenvalue of S_C will reach zero, and for $P > P_0$. For given Z , we define

$$P_0 = \sup\{\hat{P} > 0 : \text{all eigenvalues of } S_C \text{ are negative and bounded away from 0 for } P < \hat{P}\} \quad (57)$$

or equivalently,

$$P_0 = \inf\{P > 0 : \text{the largest eigenvalue of } S_C \text{ is zero}\}. \quad (58)$$

We will examine properties of P_0 in Sect. 5. In particular, in Theorem 5.5, we will see that the value of P_0 is the smallest positive, nontrivial solution to (96) below. In Lemma 5.8, we will see that for large Z , $P_0 \approx \pi^2 Z$. In fact, Fig. 1 shows that Proposition 4.6 provides an approximation of P_0/Z that is close to optimal for all positive Z other than $Z \ll 1$.

Now we show that the inverse of the linearization S_C is compact.

Lemma 4.7. *Given $Z > 0$, if $P < P_0$, then S_C is invertible and $S_C^{-1} : X \rightarrow X$ is compact.*

Proof. Assume $P < P_0$. Then 0 is not an eigenvalue of S_C . That S_C is invertible follows from the Lax-Milgram Theorem (see Proposition 6.4 for details).

First we show that S_C^{-1} is bounded. Suppose, to the contrary that it is unbounded. Then there exist sequences $(v_k) \subset \tilde{X}_C^2$ and $(w_k) \subset L^2(-1/2, 1/2)$ such that

$$S_C v_k = w_k, \quad \|v_k\|_{L^2} = 1, \quad \|w_k\|_{L^2} \leq 1/k. \quad (59)$$

Let ϕ_k satisfy $-Z\phi_k'' + \phi_k = P v_k$ with periodic boundary conditions. Then the following sequence is bounded:

$$\langle w_k, v_k \rangle_{L^2} = \langle S_C v_k, v_k \rangle_{L^2} = -\|v_k'\|_{L^2}^2 + \langle \phi_k'', v_k \rangle_{L^2}. \quad (60)$$

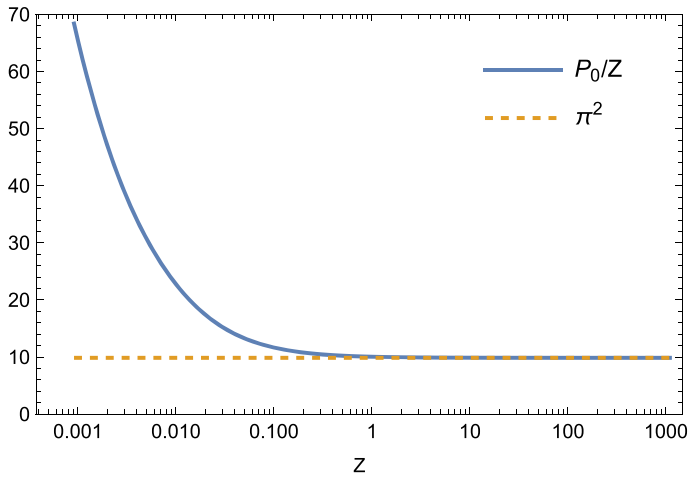


Fig. 1. The numerically calculated value of P_0/Z compared to π^2 for a wide range of Z values

The sequence $\langle \phi_k'', v_k \rangle_{L^2}$ is bounded in k due to Proposition 7.1 in Appendix A. Since the (60) as a whole is bounded, we conclude that $\|v_k'\|_{L^2}$ is bounded in k as well.

Since $\|v_k\|_{L^2}$ and $\|v_k'\|_{L^2}$ are both bounded, we conclude that (v_k) is bounded with respect to the H^1 norm. By the Banach-Alaoglu Theorem [84], there is a subsequence also called v_k which converges weakly in $H^1(-1/2, 1/2)$ and thus also in L^2 . By Morrey's inequality [37] and the Arzela-Ascoli theorem [17], $H^1(-1/2, 1/2) \subset\subset L^2(-1/2, 1/2)$, so we may assume that (v_k) converges strongly to some $v \in L^2$. Since $\|w_k\|_{L^2} \leq 1/k$, $w_k \rightarrow 0$ in $L^2(-1/2, 1/2)$. Therefore, v is a weak solution to $S_C v = 0$. Since $\lambda I - S_C$ is invertible, we conclude that $v = 0$. However, since $\|v_k\|_{L^2} = 1$, we also have $\|v\|_{L^2} = 1$, a contradiction. Therefore, S_C^{-1} is a bounded operator.

Now we show that $S_C^{-1} : L^2(-1/2, 1/2) \rightarrow L^2(-1/2, 1/2)$ is compact. To that end, suppose that $(v_k) \subset \tilde{X}_C^2$ and $(w_k) \subset L^2(-1/2, 1/2)$ such that

$$S_C v_k = w_k \quad \text{and} \quad \|w_k\|_{L^2} \leq 1. \quad (61)$$

We need to show that there (v_k) has a convergent subsequence. But this follows from the same logic as the above step. Since S_C^{-1} is a bounded operator, (v_k) is a bounded sequence in $L^2(-1/2, 1/2)$. Therefore, once again each term in (60) is bounded, so (v_k) has a weakly convergent subsequence in X^1 and a strongly convergent subsequence in X^0 . Therefore, S_C^{-1} is compact. $\square$

We can now prove the linear stability of stationary states.

Proof of Theorem 4.3. Fix $Z > 0$ and let P_0 be as described above so that if $P < P_0$, all eigenvalues of S_C are negative and bounded away from 0. Then there exists $\omega > 0$ so that for all eigenvalues λ of S_C , $\lambda \leq -\omega$. By Lemma 4.7,

S_C has compact inverse. By Lemma 4.5, S_C is self-adjoint, and therefore S_C^{-1} is also self-adjoint. By the spectral theorem [60], the eigenvectors of S_C^{-1} form an orthonormal set that spans a dense subset of $\tilde{X}_2$. Denote these eigenvectors (v_n) for $n \in \mathbb{N}$ (since $\tilde{X}_C^2$ has countable dimension, we can enumerate the eigenvectors in this way). The eigenvectors of S_C^{-1} are also eigenvectors of S_C . For each $n \in \mathbb{N}$, let λ_n be the eigenvalue of λ_n corresponding to v_n .

Suppose $u(x, t)$ solves $\partial_t u = S_C u$ in $\tilde{X}_2$ with $u(0, x) = u_0(x) \in \tilde{X}_C^2$. Since the span of (v_n) is dense in $\tilde{X}_C^2$, we may write any $u(x, t)$ as an infinite linear combination of the eigenvectors

$$u(x, t) = \sum_{n=1}^{\infty} c_n(t) v_n(x) \quad (62)$$

for coefficients $c_n : [0, \infty) \rightarrow \mathbb{R}$. Substituting this expansion into the linear evolution equation, we obtain

$$\sum_{n=1}^{\infty} c_n'(t) v_n(x) = \sum_{n=1}^{\infty} c_n(t) S_C v_n(x) = \sum_{n=1}^{\infty} \lambda_n c_n(t) v_n(x). \quad (63)$$

By the orthogonality of the eigenvectors, we conclude that the sums must agree term-by-term, so for each n , $c_n' = \lambda_n c_n$, so

$$c_n(t) = c_n(0) e^{\lambda_n t}. \quad (64)$$

Thus,

$$\begin{aligned} \|u(\cdot, t)\|_{L^2} &= \sqrt{\langle u, u \rangle_{L^2}} = \sqrt{\sum_{n=1}^{\infty} c_n^2(t)} = \sqrt{\sum_{n=1}^{\infty} c_n^2(0) e^{2\lambda_n t}} \\ &= e^{-\omega t} \sqrt{\sum_{n=1}^{\infty} c_n^2(0)} = e^{-\omega t} \|u(0, \cdot)\|_{L^2}. \end{aligned}$$

Thus, the desired result holds. $\square$

It remains to show that the full, nonlinear stability result of Theorem 4.1 holds. To this end, we consider the *nonlinear part* Ψ of F_C defined for $u \in \tilde{X}_C^2$ by

$$\Psi(u) = F_C(1 + u) - S_C u = \phi'(1/2)u' - (u\phi')', \quad (65)$$

where ϕ solves $-Z\phi'' + \phi = Pu$ with periodic boundary conditions in $(-1/2, 1/2)$. The key to our proof of nonlinear stability is showing that

the nonlinear part Ψ dominates the linear part S_C . We begin with a Lemma which gives a bound on $\Psi(u)$ in terms of u .

Lemma 4.8. *Let $\Psi : H^2([-1/2, 1/2]) \rightarrow H^1([-1/2, 1/2])$ be defined as in (65). Then there exists $C > 0$ independent of u , P , and Z such that*

$$\|\Psi(u)\|_{L^2} \leq \frac{CP}{Z} \|u\|_{L^2} \|u\|_{H^1}. \quad (66)$$

Proof. We make a direct calculation using estimates from Proposition 7.1 in Appendix A:

$$\|\Psi(u)\|_{L^2}^2 = \int_{-1/2}^{1/2} [(\phi'(1/2) - \phi'(x))u'(x) - u(x)\phi''(x)]^2 dx \quad (67)$$

$$\leq 2 \int_{-1/2}^{1/2} (\phi'(1/2) - \phi'(x))^2 (u'(x))^2 dx + 2 \int_{-1/2}^{1/2} (\phi''(x))^2 (u(x))^2 dx \quad (68)$$

$$\leq 4|\phi'(1/2)|^2 \|u'\|_{L^2}^2 + 4\|\phi'^2 u'^2\|_{L^1} + 2\|\phi''^2 u^2\|_{L^1} \quad (69)$$

$$\leq \frac{P^2}{Z^2} \|u\|_{L^1}^2 \|u'\|_{L^2}^2 + 4\|\phi'\|_{L^\infty}^2 \|u'\|_{L^2}^2 + 2\|\phi''\|_{L^4}^2 \|u\|_{L^4}^2 \quad (70)$$

$$\leq \frac{2P^2}{Z^2} \|u\|_{L^1}^2 \|u\|_{H^1}^2 + \frac{8P^2}{Z^2} \|u\|_{L^4}^4. \quad (71)$$

By Hölder's inequality, $\|u\|_{L^1} \leq \|u\|_{L^2}$. From the Gagliardo-Nirenberg inequality, there exists $C_1 > 0$ independent of u such that

$$\|u\|_{L^4} \leq C_1 \|u\|_{H^1}^{1/2} \|u\|_{L^1}^{1/2}. \quad (72)$$

Substituting (72) into (71) and letting $C^2 = 2 + 8C_1$, we obtain

$$\|\Psi(u)\|_{L^2} \leq C \frac{P}{Z} \|u\|_{L^2} \|u\|_{H^1}. \quad (73)$$

□

Our next goal is to show that if $\|u\|_{L^2}$ is small, then so is $\|\Psi(u)\|_{L^2}$. However, Lemma 4.8 is not sufficient to accomplish this because even if $\|u\|_{L^2}$ is small, $\|u\|_{H^1}$ may be large. Therefore, the following lemma shows that if $\|u(\cdot, t)\|_{L^2}$ is small for all t , then $\|u'(\cdot, t)\|_{L^2}$ does not exceed $\|u'(\cdot, 0)\|_{L^2}$.

Lemma 4.9. *Let $Z > 0$. Suppose $P < P_0$. Let $T, \varepsilon > 0$ and let u be a solution to $\partial_t u = S_C u + \Psi(u)$ in $C^1([0, T]; \tilde{X}_C^2)$. There exists $U^* > 0$ such that if $\|u'(\cdot, 0)\|_{L^2} < \varepsilon$ and $\|u(\cdot, t)\|_{L^2} < U^*$ for all $0 \leq t \leq T$, then*

$$\|u'(\cdot, t)\|_{L^2} \leq \varepsilon \text{ for all } 0 \leq t \leq T. \quad (74)$$

Proof. Write the evolution equation for u as

$$\partial_t u - u'' = -\phi'' + \Psi(u). \quad (75)$$

Square both sides and integrate to obtain

$$\|-\phi'' + \Psi(u)\|_{L^2}^2 = \int_{-1/2}^{1/2} (\partial_t u)^2 - 2(\partial_t u)u'' + (u'')^2 dx \quad (76)$$

$$= \|\partial_t u\|_{L^2}^2 + 2 \int_{-1/2}^{1/2} (\partial_t u)u' dx + \|u''\|_{L^2}^2 \quad (77)$$

$$= \|\partial_t u\|_{L^2}^2 + \frac{d}{dt} \|u'\|_{L^2}^2 + \|u''\|_{L^2}^2. \quad (78)$$

Thus,

$$\frac{d}{dt} \|u'\|_{L^2}^2 \leq \|-\phi'' + \Psi(u)\|_{L^2}^2 - \|u''\|_{L^2}^2 \quad (79)$$

$$\leq 2\|\phi''\|_{L^2}^2 + 2\|\Psi(u)\|_{L^2}^2 - \|u''\|_{L^2}^2. \quad (80)$$

From Lemma 4.8, there exists C_1 independent of u such that $\|\Psi(u)\|_{L^2} \leq C_1 \|u\|_{L^2} \|u\|_{H^1}$. Moreover, by Proposition 7.1 in Appendix A, $\|\phi''\|_{L^2} \leq 2P/Z \|u\|_{L^2}$. Since $\int_{-1/2}^{1/2} u \, dx = 0$ and $u'(\pm 1/2, t) = 0$, we may apply the Poincaré inequality to both u and u' with the optimal Poincaré constant $1/\pi$:

$$\|u\|_{L^2} \leq \frac{1}{\pi} \|u'\|_{L^2} \quad \text{and} \quad \|u'\|_{L^2} \leq \frac{1}{\pi} \|u''\|_{L^2}. \quad (81)$$

Thus,

$$\frac{d}{dt} \|u'\|_{L^2}^2 \leq 8 \frac{P^2}{Z^2} \|u\|_{L^2}^2 + 2C_1^2 \|u\|_{L^2}^2 \|u\|_{H^1}^2 - \pi^2 \|u'\|_{L^2}^2 \quad (82)$$

$$\leq 8 \frac{P^2}{Z^2} \|u\|_{L^2}^2 + 2C_1^2 \|u\|_{L^2}^2 (\|u\|_{L^2} + \|u'\|_{L^2})^2 - \pi^2 \|u'\|_{L^2}^2 \quad (83)$$

$$\leq -\left(\pi^2 - 4C_1^2 \|u\|_{L^2}^2\right) \|u'\|_{L^2}^2 + 8 \frac{P^2}{Z^2} \|u\|_{L^2}^2. \quad (84)$$

Let $U^* < \pi/(4C_1)$. Then if $\|u\|_{L^2} \leq U^*$ for all $0 < t < T$,

$$\frac{d}{dt} \|u'\|_{L^2}^2 \leq -R_1 \|u'\|_{L^2}^2 + R_2 \quad (85)$$

where

$$R_1 = \pi^2 - 4C_1^2 (U^*)^2 > \frac{\pi^2}{2} \quad \text{and} \quad R_2 = 8 \frac{P^2}{Z^2} (U^*)^2. \quad (86)$$

Let $q(t) = \|u'(\cdot, t)\|_{L^2}^2 - R_2/R_1$. Then q satisfies $q' \leq -R_1 q$. By the Grönwall's inequality, $q(t) \leq q(0)e^{-R_1 t}$. We conclude that if $q(0) < 0$, then $q(t) < 0$ for all $t > 0$. Thus, if $\|u\|_{L^2} \leq U^*$, and if

$$\|u'(\cdot, 0)\|_{L^2} < \sqrt{\frac{R_2}{R_1}}, \quad \text{then} \quad \|u'(\cdot, t)\|_{L^2} < \sqrt{\frac{R_2}{R_1}} \quad (87)$$

for all $t > 0$. Letting $U^* < \varepsilon/(4\pi)$, we have

$$\sqrt{\frac{R_2}{R_1}} < 4\pi U^* < \varepsilon, \quad (88)$$

so the desired result holds. $\square$

Proof of Theorem 4.1. Let $u = m - m_S = m - 1$. Then u solves

$$\begin{cases} \partial_t u = S_C u + \Psi(u) & -1/2 < x < 1/2, \quad t > 0 \\ u' = 0 & x = \pm 1/2, \quad t > 0 \\ u(\cdot, 0) = m_0 - 1 := u_0 & t = 0. \end{cases} \quad (89)$$

Let $S(t)$ be the semigroup generated by S_C . By Theorem 4.3, there exists $\omega > 0$ such that $\|S(t)\|_{L^2} \leq e^{-\omega t}$. Applying Duhamel's principle,

$$u(\cdot, t) = S(t)u_0 + \int_0^t S(t - \tau)\Psi(u(\tau, \cdot)) \, d\tau. \quad (90)$$

Taking the L^2 norm of both sides, we find that

$$\|u(\cdot, t)\|_{L^2} \leq e^{-\omega t} \|u_0\|_{L^2} + \int_0^t e^{\omega(\tau-t)} \|\Psi(u(\tau, \cdot))\|_{L^2} \, d\tau. \quad (91)$$

Lemma 4.8 provides an estimate for Ψ in terms of a constant C , leading to

$$\|u(\cdot, t)\|_{L^2} \leq e^{-\sigma t} \|u_0\|_{L^2} + C \int_0^t e^{\omega(\tau-t)} \|u(\tau, \cdot)\|_{L^2} \|u(\tau, \cdot)\|_{H^1} \, d\tau. \quad (92)$$

Let $\varepsilon = \omega\pi/(2C(1 + \pi))$, and let U^* be as in Lemma 4.9. Suppose that $\|u'(0, \cdot)\|_{L^2} < \varepsilon$ and $\|u(0, \cdot)\|_{L^2} < U^*$. Let

$$W = \{t \geq 0 : \|u(\tau, \cdot)\|_{L^2} \leq U^* \text{ for all } 0 \leq \tau \leq t\}. \quad (93)$$

By continuity, W is a closed interval and $0 \in W$. Thus, either $W = [0, \infty)$ or W has a positive maximum. Let $T \in W$. Then, after applying the Poincaré inequality and Lemma 4.9, for any $0 \leq t \leq T$,

$$\begin{aligned} \|u(\cdot, t)\|_{L^2} &\leq e^{-\omega t} \|u_0\|_{L^2} + C \int_0^t e^{\omega(\tau-t)} \|u(\tau, \cdot)\|_{L^2} \left(1 + \frac{1}{\pi}\right) \|u'(\tau, \cdot)\|_{L^2} \, d\tau \\ &\leq e^{-\omega t} \|u_0\|_{L^2} + C \int_0^t e^{\omega(\tau-t)} \|u(\tau, \cdot)\|_{L^2} \left(1 + \frac{1}{\pi}\right) \varepsilon \, d\tau \\ &\leq e^{-\omega t} \|u_0\|_{L^2} + \frac{\omega}{2} \int_0^t e^{\omega(\tau-t)} \|u(\tau, \cdot)\|_{L^2} \, d\tau. \end{aligned} \quad (94)$$

Therefore, by Grönwall's inequality,

$$\|u(\cdot, t)\|_{L^2} \leq \|u_0\|_{L^2} e^{-\omega t/2} \quad (95)$$

for all $0 \leq t \leq T$. Therefore, $\|u(\cdot, T)\|_{L^2} < U^*$ and so by continuity, $T \neq \max W$. Since $T \in W$ is arbitrary, we conclude that W does not have a maximum so $W = [0, \infty)$ and $\|u(\cdot, t)\|_{L^2} \leq U^*$ for all $t > 0$. That is, (95) holds for all $t \geq 0$ provided $\|u(0, \cdot)\|_{L^2} < U^*$ and $\|u'(0, \cdot)\|_{L^2} < \varepsilon$. Note that from the proof of Lemma 4.9, $U^* < \varepsilon$. Therefore, the desired result holds. $\square$

5. Traveling Waves

In this section, we show that for any $Z > 0$, there exists a number $V^* > 0$ and a smooth function $P_{TW} : (-V^*, V^*) \rightarrow \mathbb{R}$ such that if $P = P_{TW}(V)$, then there exists a traveling wave solution to the model B with velocity V and center $c = Vt$. This family of traveling wave solutions parameterized by V bifurcates from the family of stationary states at $V = 0$ and $P = P_{TW}(0) = P_0$. As we show, this bifurcation is of the type illustrated in Fig. 2.

We will see that, for a given Z , the bifurcation occurs at a positive solution P_0 to

$$\begin{aligned} \tanh\left(\frac{\sqrt{1-P_0}}{2\sqrt{Z}}\right) &= P_0 \frac{\sqrt{1-P_0}}{2\sqrt{Z}} \quad \text{or equivalently} \\ \tan\left(\frac{\sqrt{P_0-1}}{2\sqrt{Z}}\right) &= P_0 \frac{\sqrt{P_0-1}}{2\sqrt{Z}}. \end{aligned} \quad (96)$$

We write (96) in two forms to emphasize that P_0 be greater than or less than 1. Of course $P_0 = 1$ is a trivial solution, but we shall see that this solution does not correspond to a bifurcation, so we are interested in nontrivial solutions to (96). In fact, for large enough Z , there are infinitely many nontrivial positive solutions to (96), indicating that there are infinitely many bifurcations and infinitely many families of traveling waves. However, only one of these families of traveling waves is exponentially stable in small velocity, and this family corresponds to the smallest positive nontrivial solution to (96), and so when we write P_0 , we refer to this solution. In the proof of Lemma 5.8, we see that if $Z < 1/12$, then $P_0 < 1$ and if $Z > 1/12$, then $P_0 > 1$. If $Z = 1/12$, then $P_0 = 1$ is a degenerate root of (96) and the transversality condition in the Crandall-Rabinowitz Theorem used in Theorem 5.5 to prove the existence of bifurcation is not satisfied. Future study will be required for the $Z = 1/12$ case.

Remark 5.1. We note that the presence of the implied bifurcation was first discovered in [79,80] where the structure of the associated bifurcation for both model A and model B was identified as well by using formal expansions. Here, in the framework of model B, we complement this earlier study not only by a rigorous analysis of the existence of the traveling wave solutions but also by providing a global analysis of the corresponding bifurcation.

Observe first that $m(x, t)$ is a traveling wave solution with velocity V in model B if $m(x, t) = m_{TW}(x - Vt)$ where m_{TW} satisfies

$$\begin{cases} m_{TW}'' + Vm_{TW}' - (m_{TW}\phi_{TW}')' = 0 & -1/2 < x < 1/2 \\ m_{TW}'(\pm 1/2) = 0 \end{cases} \quad \text{with} \quad \begin{cases} -Z\phi_{TW}'' + \phi_{TW} = P(V)m_{TW} & -1/2 < x < 1/2 \\ \phi_{TW}(1/2) = \phi_{TW}(-1/2) \\ \phi_{TW}'(1/2) = \phi_{TW}'(-1/2) = V. \end{cases} \quad (97)$$

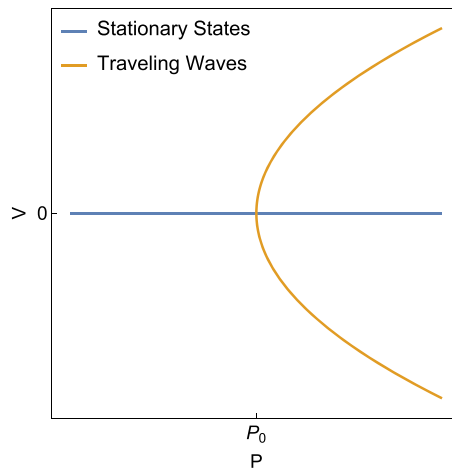


Fig. 2. This diagram shows the pitchfork bifurcation from stationary states to traveling waves which is structurally the same in all three models

We further observe that a solution to this equation is $m_{TW} = \Lambda e^{\phi_{TW} - Vx}$ for any $\Lambda \in \mathbb{R}$. The value of $\Lambda = \Lambda(V)$ can be determined by the provision that $\int_{-1/2}^{1/2} m \, dx = 1$, and the value of ϕ_{TW} therefore satisfies

$$\begin{cases} -Z\phi_{TW}'' + \phi_{TW} = P(V)\Lambda(V)e^{\phi_{TW}(x) - Vx} & -1/2 < x < 1/2 \\ \phi_{TW}(-1/2) = \phi_{TW}(1/2) \\ \phi_{TW}'(-1/2) = \phi_{TW}'(1/2) = V. \end{cases} \quad (98)$$

Note that (98) has three boundary conditions: not only must ϕ_{TW} satisfy periodic boundary conditions, but also $\phi_{TW}'(\pm 1/2) = V$. Thus, $P(V)$ is selected so that ϕ_{TW} can satisfy this extra condition.

Solutions to (98) may be approximated asymptotically in small V as shown in the following Lemma:

Lemma 5.2. *Let $Z > 0$ with $Z \neq 1/12$. Suppose that m_{TW} , ϕ_{TW} and $P(V)$ solve (97). In small V , $P(V)$ and m_{TW} have the asymptotic forms*

$$P(V) = P_0 + V^2 P_2 + O(V^4) \quad (99)$$

$$m_{TW}(x) = 1 + V m_1(x) + V^2 m_2(x) + O(V^3), \quad (100)$$

where

- P_0 is a nontrivial solution to (96)
- P_2 is given by

$$P_2 = \frac{P_0(6P_0^6 - 15P_0^5 - 3P_0^4(56Z - 5) + P_0^3(514Z - 6) - 1044P_0^2Z + 72P_0Z(55Z - 1) + 5280Z^2)}{288(P_0 - 1)^4(P_0^2 - 12Z)}. \quad (101)$$

• m_1 is given by

$$m_1(x) = \frac{x}{P_0 - 1} - \frac{P_0 \sqrt{\frac{P_0^3 - P_0^2 + 4Z}{(P_0 - 1)P_0^2}} \sin\left(\frac{\sqrt{P_0 - 1}x}{\sqrt{Z}}\right)}{2P_0 - 2} \quad (102)$$

• m_2 is given by

$$\begin{aligned} m_2(x) = & A + Bx^2 + (C + dx^2) \cos\left(\frac{\sqrt{P_0 - 1}}{\sqrt{Z}}x\right) \\ & + Ex \sin\left(\frac{\sqrt{P_0 - 1}}{\sqrt{Z}}x\right) + F \cos\left(\frac{2\sqrt{P_0 - 1}}{\sqrt{Z}}x\right) \end{aligned} \quad (103)$$

with

$$A = \frac{-6P_0Z + P_0 + 24Z - 1}{24(P_0 - 1)^4} \quad (104)$$

$$B = \frac{12 - 12P_0}{24(P_0 - 1)^4} \quad (105)$$

$$C = \frac{(P_0(28Z - 3) + 3P_0^2 - 60Z) \sqrt{\frac{P_0^3 - P_0^2 + 4Z}{Z}}}{96(P_0 - 1)^4} \quad (106)$$

$$D = -\frac{P_0 \sqrt{\frac{P_0^3 - P_0^2 + 4Z}{Z}}}{8(P_0 - 1)^3} \quad (107)$$

$$E = \frac{(4 - 3P_0) \sqrt{Z} \sqrt{\frac{P_0^3 - P_0^2 + 4Z}{Z}}}{8(P_0 - 1)^{7/2}} \quad (108)$$

$$F = \frac{(3 - 4P_0)(P_0^3 - P_0^2 + 4Z)}{48(P_0 - 1)^4}. \quad (109)$$

Proof. We first introduce expansions for $\Lambda(V)$ and ϕ_{TW} :

$$\Lambda(V) = \Lambda_0 + V^2 \Lambda_2 + O(V^4), \quad (110)$$

$$\phi_{TW}(x) = \phi_0(x) + V\phi_1(x) + V^2\phi_2(x) + V^3\phi_3(x) + O(V^4). \quad (111)$$

Observe that neither the expansion for P (99) nor the expansion for Λ (110) have terms that are of odd order in V . This is because we expect symmetry in traveling wave solutions with respect to the sign of V . Therefore, $P(V)$ and $\Lambda(V)$ are even functions of V and for $m_{TW}(x)$ and $\phi_{TW}(x)$, the transformation $V \mapsto -V$ is equivalent to $x \mapsto -x$.

Since $m_{TW}(x) = \Lambda(V)e^{\phi_{TW}(x) - Vx}$, we may expand the exponential to obtain $1 = \Lambda_0 e^{\phi_0}$, $m_1(x) = \Lambda_0 e^{\phi_0}(\phi_1 - x)$ and

$$m_2(x) = \Lambda_2 e^{\phi_0} + \frac{1}{2} \Lambda_0 e^{\phi_0} (x^2 - 2x\phi_1 + \phi_1^2 + 2\phi_2). \quad (112)$$

We conclude that ϕ_0 is constant and $\Lambda_0 = e^{-\phi_0}$. Substituting the expansion (111) into (98) and comparing terms of like order in V , we obtain $\phi_0 = P_0$ in zeroth order and the following differential equations in first through third order:

$$\begin{cases} -Z\phi_1'' + (1 - P_0)\phi_1 = -P_0x & -1/2 < x < 1/2 \\ \phi_1(1/2) = \phi_1(-1/2) \\ \phi_1'(1/2) = \phi_1'(-1/2) = 1 \end{cases} \quad (113)$$

$$\begin{cases} -Z\phi_2'' + (1 - P_0)\phi_2 = P_2 + P_0e^{P_0}\Lambda_2 + \frac{P_0}{2}(\phi_1 - x)^2 & -1/2 < x < 1/2 \\ \phi_2(1/2) = \phi_2(-1/2) \\ \phi_2'(1/2) = \phi_2'(-1/2) = 0 \end{cases} \quad (114)$$

$$\begin{cases} -Z\phi_3'' + (1 - P_0)\phi_3 = (\phi_1 - x) \\ \quad \left(P_0(\Lambda_2e^{P_0} + \phi_2) + P_2 \right) \\ \quad + \frac{1}{6}P_0(\phi_1 - x)^3 & -1/2 < x < 1/2 \\ \phi_3(1/2) = \phi_3(-1/2) \\ \phi_3'(1/2) = \phi_3'(-1/2) = 0. \end{cases} \quad (115)$$

If $P_0 = 1$, then the solution to (113) is

$$\phi_1 = \frac{x^3}{6Z} - \frac{x}{24Z} + \beta \quad (116)$$

for some $\beta \in \mathbb{R}$. To satisfy $\phi_1'(\pm 1/2) = 1$, we must have $Z = 1/12$. This is a contradiction, so $P_0 \neq 1$.

If $P_0 \neq 1$, then the solution to (113) is

$$\phi_1(x) = \frac{P_0x}{P_0 - 1} - \frac{1}{2} \frac{P_0}{P_0 - 1} \csc\left(\frac{\sqrt{P_0 - 1}}{2\sqrt{Z}}\right) \sin\left(\frac{\sqrt{P_0 - 1}}{2\sqrt{Z}}x\right). \quad (117)$$

In order to satisfy the additional condition $\phi_1'(\pm 1/2) = 1$, P_0 must solve (96). Therefore, (102) is obtained as $m_1(x) = \phi_1(x) - x$.

In (114), Λ_2 is determined by the condition that

$$\frac{d^2}{dV^2} \int_{-1/2}^{1/2} m_{TW}(x) dx = \frac{d^2}{dV^2} \int_{-1/2}^{1/2} \Lambda(V)e^{\phi_{TW} - Vx} dx = 0. \quad (118)$$

The value of Λ_2 is

$$\Lambda_2 = -e^{-P_0}P_2 - \frac{e^{-P_0}(3P_0^2 - 60Z + 2)}{48(P_0 - 1)^2}. \quad (119)$$

The solution to (114) can be found using elementary methods and has the form

$$\begin{aligned} \phi_2(x) = & P_2 + a_0 + a_2x^2 + (b_0 + b_2x^2) \cos\left(\frac{\sqrt{P_0 - 1}}{\sqrt{Z}}x\right) \\ & + c_1x \sin\left(\frac{\sqrt{P_0 - 1}}{\sqrt{Z}}x\right) + d_0 \cos\left(\frac{2\sqrt{P_0 - 1}}{\sqrt{Z}}x\right) \end{aligned} \quad (120)$$

where

$$a_0 = \frac{P_0^2(1 - 30Z) + P_0(48Z - 1)}{24(P_0 - 1)^4} \quad (121)$$

$$a_2 = -\frac{P_0}{2(P_0 - 1)^3} \quad (122)$$

$$b_0 = \frac{(P_0(28Z - 3) + 3P_0^2 - 60Z)\sqrt{P_0^3 - P_0^2 + 4Z}}{96(P_0 - 1)^4\sqrt{Z}} \quad (123)$$

$$b_2 = -\frac{P_0\sqrt{P_0^3 - P_0^2 + 4Z}}{8(P_0 - 1)^3\sqrt{Z}} \quad (124)$$

$$c_1 = \frac{P_0\sqrt{P_0^3 - P_0^2 + 4Z}}{8(P_0 - 1)^{7/2}} \quad (125)$$

$$d_0 = \frac{-4P_0Z - P_0^4 + P_0^3}{48(P_0 - 1)^4}. \quad (126)$$

Note that the only dependence on P_2 in ϕ_2 is the leading term—none of the other coefficients depend on P_2 . Therefore, we write $\phi_2 = P_2 + \tilde{\phi}_2$, where $\tilde{\phi}_2$ is independent of P_2 . We similarly write $\Lambda_2 = -e^{-P_0}P_2 + \tilde{\Lambda}_2$.

In third order, we do not need to find an explicit solution ϕ_3 . Instead, we divide the right hand side of the differential equation in (115) to separate terms that explicitly depend on P_2 from those that do not:

$$-Z\phi_3'' + (1 - P_0)\phi_3 = P_2f(x) + g(x), \quad (127)$$

where $f(x) = \phi_1 - x$ and $g(x) = P_0(\phi_1 - x)\left(e^{P_0}\tilde{\Lambda}_2 + \tilde{\phi}_2\right) + \frac{1}{6}P_0(\phi_1 - x)^3$. The three boundary conditions that ϕ_3 must satisfy (periodic boundary conditions with $\phi_3'(\pm 1/2) = 0$) determine P_2 , which we show as follows. Let $U(x) = \sin\left(x\sqrt{P_0 - 1}/\sqrt{Z}\right)$. Then

$$\int_{-1/2}^{1/2} (P_2f(x) + g(x))U(x) dx = \int_{-1/2}^{1/2} (-Z\phi_3'' + (1 - P_0)\phi_3)U(x) dx \quad (128)$$

$$\begin{aligned} &= -Z\phi_3'U(x)\Big|_{-1/2}^{1/2} + Z\phi_3U'\Big|_{-1/2}^{1/2} \\ &\quad + \int_{-1/2}^{1/2} \phi_3(-ZU'' + (1 - P_0)U) dx \end{aligned} \quad (129)$$

$$= 0. \quad (130)$$

Therefore, we may explicitly calculate

$$\begin{aligned} P_2 &= -\frac{\int_{-1/2}^{1/2} g(x)U(x) dx}{\int_{-1/2}^{1/2} f(x)U(x) dx} \\ &= \frac{P_0(6P_0^6 - 15P_0^5 - 3P_0^4(56Z - 5) + P_0^3(514Z - 6) - 1044P_0^2Z + 72P_0Z(55Z - 1) + 5280Z^2)}{288(P_0 - 1)^4(P_0^2 - 12Z)}. \end{aligned} \quad (131)$$

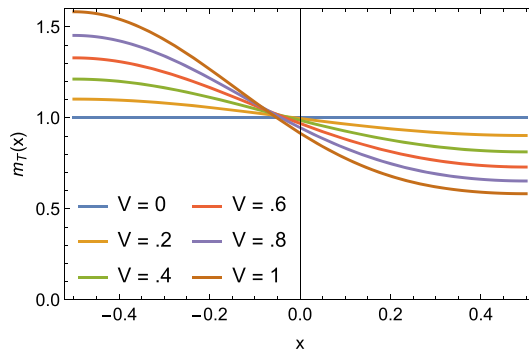


Fig. 3. Myosin density of traveling waves for the smallest nontrivial solution P_0 to (96). Traveling waves with low velocity have nearly constant myosin density ($m_{TW} \approx 1$), but traveling waves with higher velocity are increasingly asymmetric

Therefore, substituting (131) into (120), and then into (112), we obtain (103). $\square$

Remark 5.3. The existence of this bifurcation, demonstrated for the 1D problem in [79, 80], was shown for the 2D problem in [86] (see also [87]). The value of P_2 in (101) is identical to the result given in [80] (see Appendix D).

Remark 5.4. Lemma 5.2 gives the asymptotic form of traveling wave solutions if they exist, but it does not prove that such solutions exist. To prove existence, we have Theorem 5.5 below. In fact, many such traveling wave solutions exist, each corresponding to a different solution to (96). Lemma 5.2 holds for any of these solutions (other than the trivial solution $P_0 = 1$), but going forward, we reserve P_0 to refer to the smallest nontrivial solution to (96).

A plot of the asymptotic approximation of $m_{TW}(x)$ for the smallest nontrivial solution P_0 to (96) given by Lemma 5.2 for several values of V is given in Fig. 3.

We can now prove the following result (see also [80]):

Theorem 5.5. *Let $Z > 0$ with $Z \neq 1/12$ and suppose P_0 is the smallest positive nontrivial solution to (96). Then there exists $V^* > 0$ and a continuous function $P_{TW} : (-V^*, V^*) \rightarrow \mathbb{R}$ such that for each $V \in (-V^*, V^*)$, there exists a family traveling wave solutions $(m_{TW}, \phi_{TW}, Vt + c_0)$ of velocity V to (34)–(39) with $P = P_{TW}(V)$ and $c_0 \in \mathbb{R}$. Moreover, $P_{TW}(0) = P_0$ and m_{TW} as a function of V is continuously differentiable function from $(-V^*, V^*)$ to $H^2(-1/2, 1/2)$.*

The parameter V^* in Theorem 5.5 is the (not explicitly known) largest velocity for which traveling waves must exist. That is, the bifurcation of stationary solutions to traveling waves is a strictly local result in a neighborhood of $(m, V) = (1, 0)$. The main tool to prove this theorem will be the CR theorem [25], which we quote in Appendix B.

Essentially, the CR theorem gives conditions under which an equation of the form $\mathcal{F}(x, t) = 0$ has two families of solutions: a *trivial branch* where $x = 0$ and t parameterizes the family, and a *nontrivial branch* where x and t are both

parameterized by a new parameter s , and the two families meet at $(x, t) = (0, 0)$. In Theorem 5.5, the trivial branch corresponds to the stationary homogeneous solution $m = 1$ for any value of P . The nontrivial branch corresponds to the traveling wave solutions parameterized by their velocity and with activity parameter $P = P_{TW}(V)$. The two families of solutions meet at $P = P_{TW}(0) = P_0$ satisfying (96).

Proof of Theorem 5.5. Given $Z > 0$ with $Z \neq 1/12$, let $P = P_0$ be a nontrivial solution to (96). Let

$$X = \left\{ \mu \in H^2(-1/2, 1/2) : \mu'(-1/2) = \mu'(1/2) = 0, \int_{-1/2}^{1/2} \mu(x) dx = 0 \right\} \quad (132)$$

and

$$Y = \left\{ s \in L^2(-1/2, 1/2) \times \mathbb{R} : \int_{-1/2}^{1/2} s(x) dx = 0 \right\}. \quad (133)$$

Define $\mathcal{F} : X \times \mathbb{R} \rightarrow Y$ by

$$\mathcal{F}(\mu, \tau) = \mu'' + \phi'(1/2)\mu' - ((\mu + 1)\phi')', \quad (134)$$

where ϕ satisfies $-Z\phi'' + \phi = (P_0 + \tau)(1 + \mu)$ with periodic boundary conditions in $(-1/2, 1/2)$. Observe that

$$m(x, t) = \mu(x - \phi'(1/2)t, t) + 1 \quad (135)$$

is a traveling wave solution to (17) if and only if $F(\mu, \tau) = 0$. If $\phi'(1/2) = 0$, then m is a stationary solution, i.e., a traveling wave with velocity 0.

We will show that $\mathcal{F}$ satisfies the properties required for the validity of the CR theorem, see Appendix B.

- It is clear that $\mathcal{F}(0, \tau) = 0$ for all τ .
- It is also clear that $\mathcal{F}$ is twice continuously differentiable.
- The linearization of $\mathcal{F}$ in μ at $(\mu, \tau) = (0, 0)$ is

$$0 = D_\mu \mathcal{F}(0, 0)u = u'' - \psi'' \quad (136)$$

where ψ satisfies $-Z\psi'' + \psi = P_0 u$ with periodic boundary conditions on $(-1/2, 1/2)$. Note that this is the operator S_C in (48). To show that the third hypothesis of the CR theorem is satisfied, we need to show two things:

- There exists a unique (up to multiplicative constant) nonzero solution $u_0 \in X$ to $D_\mu \mathcal{F}(0, 0)u_0 = 0$, and
- there exists a co-dimension one subspace X' of X such that if $w \in X'$, then there exists a solution $u \in X$ to $D_\mu \mathcal{F}(0, 0)u = w$.

First we show (i). Consider the function

$$u_0(x) = \frac{\sqrt{Z} \operatorname{sech}\left(\frac{\sqrt{1-P_0}}{2\sqrt{Z}}\right) \sinh\left(\frac{\sqrt{1-P_0}x}{\sqrt{Z}}\right)}{(1-P_0)^{3/2}} + \frac{x}{P_0 - 1}. \quad (137)$$

Observe that $u'_0(\pm 1/2) = 0$ and $\int_{-1/2}^{1/2} u_0(x) dx = 0$, so $u_0 \in X$. Moreover, one may check that provided P_0 satisfies (96), then $D_\mu \mathcal{F}(0, 0)u_0 = 0$. Now suppose that u_1 and u_2 are both nonzero solutions to (136). Let ψ_i solve $-Z\psi_i'' + \psi_i = P_0 u_i$ with periodic boundary conditions on $(-1/2, 1/2)$ for $i = 1, 2$. Observe that for each i , since the second derivatives of u_i and ψ_i are equal, we have

$$u_i - \psi_i = \alpha_i x - \beta_i \quad (138)$$

We also observe that

$$\int_{-1/2}^{1/2} \psi_i dx = P_0 \int_{-1/2}^{1/2} u_i dx + Z \int_{-1/2}^{1/2} \psi_i'' dx = 0. \quad (139)$$

Therefore, $\beta_1 = \beta_2 = 0$. Now, suppose for some i , $\psi'_i(\pm 1/2) = 0$. Then $\alpha_i = 0$ and $u_i = \psi_i$. Thus, u_i satisfies $-Zu_i'' + u_i = P_0 u_i$ with periodic boundary conditions. We conclude that u_i is an eigenvector of the second derivative operator with eigenvalue $(1 - P_0)/Z$. The eigenvalues of the second derivative operator on X are $-n^2\pi^2$ for positive integers n . But since P_0 and Z must satisfy (96) and $P_0 \neq 1$, it is clear that $(1 - P_0)/Z \neq -n^2\pi^2$, so we have arrived at a contradiction. Therefore, $\psi'_i(\pm 1/2) \neq 0$, and we may assume without loss of generality that the u_i are scaled such that $\psi'_i(\pm 1/2) = 1$ for $i = 1, 2$, so $\alpha_i = -1$. Then $u_i = \psi_i - x$. Let $w = u_1 - u_2$. Then

$$w = (\psi_1 - x) - (\psi_2 - x) = \psi_1 - \psi_2. \quad (140)$$

Thus, w satisfies $-Zw'' + w = P_0 w$ with periodic boundary conditions on $(-1/2, 1/2)$. Once again, since $(1 - P_0)/Z \neq -n^2\pi^2$, the only solution is $w = 0$. We conclude that $u_1 = u_2$ and u_0 is unique up to a multiplicative constant.

Now we show (ii). Observe that since ψ satisfies $\psi'' = (P_0 u - \psi)/Z$, we may write

$$D_m \mathcal{F}(0, 0)u = u'' - \frac{P_0}{Z}u + \frac{1}{Z}\psi. \quad (141)$$

We may abstract this operator as $D_m \mathcal{F}(0, 0) = B + K$ where $Bu = u'' - (P_0/Z)u$ and $K : u \mapsto \psi/Z$. We make two observations. First, K is a bounded operator with respect to the H^2 norm and a compact operator with respect to the L^2 norm. However, since we will only apply K to $u \in X \subset H^2(-1/2, 1/2)$, we may restrict to domain of K to X , and then by the Rellich–Kondrachov Theorem, it is a compact operator in the H^2 norm. Second, the operator B is invertible on X and its inverse B^{-1} is bounded. Therefore, we may write

$$I - B^{-1}D_m \mathcal{F}(0, 0) = I - B^{-1}(B + K) = -B^{-1}K \quad (142)$$

and

$$I - D_\mu \mathcal{F}(0, 0)B^{-1} = I - (B + K)B^{-1} = -KB^{-1}. \quad (143)$$

Since B^{-1} is bounded and K is compact, the operators $B^{-1}K$ and KB^{-1} are compact in H^2 . We conclude that $D_\mu \mathcal{F}(0, 0)$ is a Fredholm operator on X . We

recall that the index of a Fredholm operator is the difference between the dimension of its kernel and the codimension of its range. We also recall that the index of a self-adjoint operator is zero. Since B and K are both self-adjoint over $L^2([-1/2, 1/2])$, so is $D_\mu \mathcal{F}(0, 0)$. Therefore, the codimension of the range of $D_\mu \mathcal{F}(0, 0)$ is equal to the dimension of the kernel, which we have just proved is 1.

- Finally, we must show that $D_{\mu s} \mathcal{F}(0, 0)u_0$ is not in the range of the operator $D_\mu \mathcal{F}(0, 0)$. Observe that since $D_\mu \mathcal{F}(0, 0)$ is self-adjoint, its image is orthogonal to its kernel. That is, for any $u \in X$, $\langle D_\mu \mathcal{F}(0, 0)u, u_0 \rangle_{L^2} = 0$. It is therefore sufficient to show that $\langle D_{\mu s} \mathcal{F}(0, 0)u_0, u_0 \rangle_{L^2} \neq 0$. The mixed second derivative is

$$D_{\mu s} \mathcal{F}(0, 0)u = u'' - \tilde{\psi}'' \quad (144)$$

where $\tilde{\psi}$ satisfies $-Z\tilde{\psi}'' + \tilde{\psi} = u$. Therefore, $\tilde{\psi} = \psi/P_0$. We conclude that

$$\begin{aligned} \langle D_{\mu s} \mathcal{F}(0, 0)u_0, u_0 \rangle_{L^2} &= \int_{-1/2}^{1/2} \left(u_0'' - \frac{\psi_0''}{P_0} \right) u_0 \, dx \\ &= \int_{-1/2}^{1/2} \left(1 - \frac{1}{P_0} \right) u_0 u_0'' \, dx = - \int_{-1/2}^{1/2} \left(1 - \frac{1}{P_0} \right) (u_0')^2 \, dx. \end{aligned}$$

Since $P_0 \neq 1$, $\langle D_{\mu s} \mathcal{F}(0, 0)u_0, u_0 \rangle_{L^2} \neq 0$ (it is also interesting to note that this is why bifurcation does not occur when $P_0 = 1$).

Since we now checked that all the hypotheses of the CR theorem hold in a neighborhood of $(\mu, \tau) = (0, 0)$, the only solutions to $\mathcal{F}(\mu, \tau) = 0$ are $\mu = 0$ plus a smooth family of solutions $(\mu(s), \tau(s))$ parameterized by s in some small interval $(-s^*, s^*)$ with $\mu(s) \not\equiv 0$. Moreover, these two families of solutions meet at $(0, 0)$. Indeed, let $m_{TW}(s) = 1 + \mu(s)$ and $P_{TW}(s) = P_0 + \tau(s)$. Moreover, $m_{TW} : (-s^*, s^*) \rightarrow H^2(-1/2, 1/2)$ is continuously differentiable.

Since all solutions to $\mathcal{F} = 0$ are traveling waves with some velocity (or stationary solutions if the velocity is zero), it only remains to show that $m_{TW}(s)$ and $P_{TW}(s)$ may be reparameterized (at least locally near $s = 0$) by velocity. Let $V(s)$ be the velocity of $m_{TW}(s)$. It is sufficient to show that $V'(0) \neq 0$. Let $\phi_{TW}(s)$ satisfy $-Z\phi_{TW}'' + \phi_{TW} = P_{TW}(s)m_{TW}(s)$ with periodic boundary conditions on $(-1/2, 1/2)$. Then $V(s) = \partial_x \phi_{TW}(s)|_{x=1/2}$. Therefore, $V'(0)$ is $\psi'(1/2)$ where ψ solves

$$\begin{aligned} -Z\psi'' + \psi &= \frac{\partial}{\partial s} P_{TW}(s)m_{TW}(s) \Big|_{s=0} = \frac{\partial P_{TW}}{\partial s}(0)m_{TW}(0) \\ &+ P_{TW}(0) \frac{\partial m_{TW}}{\partial s}(0) = \frac{\partial P_{TW}}{\partial s}(0) + P_0 \frac{\partial \mu}{\partial s}(0), \end{aligned} \quad (145)$$

with periodic boundary conditions. Since $\mathcal{F}$ is twice differentiable, $\mu(s)$ is also continuously differentiable and $\mu'(0)$ spans the null space of $\mathcal{F}_\mu(0, 0)$. Without loss of generality, we may assume that $\mu(s)$ is parameterized such that

$$\frac{d\mu}{ds}(0) = \mu_0. \quad (146)$$

Let ψ_0 solve $-Z\psi_0'' + \psi_0 = P_0\mu_0$ with periodic boundary conditions on $(-1/2, 1/2)$. Then ψ_0 and ψ differ by a constant, so $V'(0) = \psi'_0(1/2)$. We may explicitly

calculate $\psi'_0(1/2) = 1$, so $V'(0) \neq 0$. Therefore, we may smoothly reparametrize m_{TW} and P_{TW} by V for V in some small interval $(-V^*, V^*)$ such that $m_{TW} : (-V^*, V^*) \rightarrow \mathbb{H}^2(-1/2, 1/2)$ is continuously differentiable. $\square$

Remark 5.6. As a result of Theorem 5.5, the map $m_{TW} : (-V^*, V^*) \rightarrow \mathbb{H}^2(-1/2, 1/2)$ is continuously differentiable. This means that $m'_{TW} : (-V^*, V^*) \rightarrow H^1(-1/2, 1/2)$ is also continuously differentiable. Since ϕ_{TW} is a bounded linear function of m_{TW} , it is also a continuously differentiable function from $(-V^*, V^*)$ to $H^2(-1/2, 1/2)$. Thus, we can say that each of m_{TW} , m'_{TW} , m''_{TW} , ϕ_{TW} , ϕ'_{TW} , and ϕ''_{TW} are all continuously differentiable as functions from $(-V^*, V^*)$ to $L^2(-1/2, 1/2)$.

Remark 5.7. While Theorem 5.5 proves the existence of a bifurcating branch of traveling wave solutions that meets the branch of stationary solutions at $P = P_0$, the smallest positive nontrivial solution to (96), the exact same proof would prove the existence of bifurcating branch of traveling waves emerging from any other positive nontrivial solution to (96). Our interest in the “first” branch of traveling wave solutions arises because this is the only family that is exponentially stable (as we shall see in Sect. 6). Thus, among all families of traveling waves, the notations m_{TW} , ϕ_{TW} , and P_{TW} are used to denote this “first” family.

Given $Z > 0$, the condition (96) satisfied by P_0 has (potentially) infinitely many solutions. Therefore, Theorem 5.5 proves the existence of not just one, but infinitely many families of traveling wave solutions, each bifurcating from the stationary solution for a different solution P to (96). In Sect. 4, we observed that for $P/Z < \pi^2$, the eigenvalues of the linearization S_C of model C about the stationary solution are all negative. In the proof of Theorem 5.5, we observe that if P satisfies (96), S_C has a zero eigenvalue. We conclude that as P/Z increases from π^2 , each solution of (96) corresponds to one of the eigenvalues of S_C becoming positive. Therefore, we conjecture that for all families of traveling waves except those bifurcating from the smallest solution of (96), the linearization of model C about these traveling waves has some positive eigenvalues, and therefore these traveling waves are unstable. The only traveling wave solutions that may be stable are those bifurcating from the smallest solution to (96). Therefore, when using the notation P_0 , we refer to this value. The following lemma shows the existence of this smallest solution and provides the illuminating estimate that, for large Z , $P_0/Z \geq \pi^2$ with equality in the limit $Z \rightarrow \infty$.

Lemma 5.8. *Suppose that $P_0 = P_0(Z)$ is the smallest nontrivial positive solution to (96) whenever such a solution exists. Then $P_0(Z)$ exists for all $Z > 0$ except $Z = 1/12$ and in large Z , $P_0(Z)$ expands as*

$$P_0(Z) = \pi^2 Z + 1 - \frac{8}{\pi^2} + O(1/Z). \quad (147)$$

Proof. First suppose $0 < Z < 1/12$. Let $v = \sqrt{1 - P_0}/(2\sqrt{Z})$. Then v and Z satisfy

$$k_1(z) := \frac{v - \tanh(v)}{4v^3} = Z. \quad (148)$$

It is easy to show that k_1 is continuous on $(0, \infty)$, $\lim_{v \rightarrow 0^+} k_1(v) = 1/12$, $\lim_{v \rightarrow \infty} k_1(v) = 0$, and k_1 is monotonically decreasing. Thus, for any $Z \in (0, 1/12)$, there exists a unique $v \in (0, \infty)$ satisfying (148). Therefore, $P_0(Z) = \tanh(v)/v \in (0, 1)$ is uniquely determined.

Now suppose $Z > 1/12$. Write (96) as

$$\tan\left(\frac{\sqrt{P_0 - 1}}{2\sqrt{Z}}\right) = \frac{P_0\sqrt{P_0 - 1}}{2\sqrt{Z}} \quad (149)$$

and let $w = \sqrt{P_0 - 1}/(2\sqrt{Z})$. Then w and Z satisfy

$$k_2(w) = \frac{\tan(w) - w}{4w^3} = Z. \quad (150)$$

Similarly to k_1 , it is easy to show that k_2 is continuous on $(0, \pi/2)$, $\lim_{w \rightarrow 0^+} k_2(w) = 1/12$, $\lim_{w \rightarrow \pi/2^-} k_2(w) = \infty$, and k_2 is monotonically increasing on $(0, \pi/2)$. Thus, for any $Z \in (1/12, \infty)$, there exists a unique $w \in (0, \pi/2)$ satisfying (150). Therefore, $P_0(Z) = \tan(w)/w \in (1, \infty)$. Thus, for all positive Z other than $Z = 1/12$, (96) has a smallest positive solution other than 1. It should be noted that using a similar line of reasoning, we may show that (150) has a unique solution in each interval of the form $(n\pi/2, (n+2)\pi/2)$ for $n > 0$ odd. These correspond to the other (larger than P_0) solutions to (96) referenced above.

As $Z \rightarrow \infty$ the corresponding solution w to $k_2(w) = Z$ approaches $\pi/2$. Therefore, we expand w in large Z as $w(Z) = \pi/2 + w_1/Z + w_2/Z^2 + O(1/Z^3)$. We expand (150) in large Z and compare terms of like order in Z to obtain $w = \pi/2 - 2/(Z\pi^3) + O(1/Z^2)$. Finally, using $P_0 = 1 + 4Zw^2$, we have

$$P_0 = \pi^2 Z + \left(1 - \frac{8}{\pi^2}\right) + O(1/Z). \quad (151)$$

□

6. Non-Self-Adjoint Spectral Analysis for Nonlinear Stability of Traveling Waves

In this section we study the nonlinear stability of traveling wave solutions to Model B. As shown in Theorem 5.5, traveling wave solutions of velocity V sufficiently small exist provided P has the prescribed value $P_{TW}(V)$. Such a traveling wave solution has the form $m(x, t) = m_{TW}(x - Vt)$ with $c = c_0 + Vt$ where m_{TW} is a stationary solution to model C. For ease of analysis we will study the stability of these solutions in the framework of model C which, as described in Sect. 4, implies stability “up to shifts” of traveling wave solutions to model B.

As in Sect. 4, we describe model C by the dynamical system $\partial_t m = F_C(m)$ with F_C given by (46). Let m_{TW} be the family of traveling wave solutions to (46) guaranteed by Theorem 5.5, and let P_{TW} be the corresponding family of activity parameters, both families parameterized by V (that is, m_{TW} is the branch of solutions bifurcating from the smallest positive nontrivial solution to (96)). The main result of this section is the following theorem about the exponential stability of m_{TW} :

Theorem 6.1. *There exist $V^*, Z^* > 0$ such that if $|V| < V^*$ and $Z > Z^*$, then the traveling wave m_{TW} is exponentially stable in the sense that there exists $\varepsilon, r, M > 0$ (depending on V and Z) such that if $m(\cdot, t)$ is a solution to (46) with $P = P_{TW}(V)$ and $m(\cdot, 0) = m_0 \in H^1(-1/2, 1/2)$ satisfying*

$$\|m_{TW} - m_0\|_{H^1} < \varepsilon, \quad (152)$$

then

$$\|m(\cdot, t) - m_{TW}\|_{L^2} \leq \|m_{TW} - m_0\|_{L^2} e^{-rt}. \quad (153)$$

While Theorem 6.1 is written to emphasize that traveling waves are stable if their velocity is small ($|V| < V^*$) and $P = P_{TW}(V)$. However, an equivalent way to write Theorem 6.1 would state that traveling waves are stable if the activity rate P is close to P_0 (more specifically, $P \in P_{TW}(-V^*, V^*)$) and velocity V satisfies $P = P_{TW}(V)$.

To prove Theorem 6.1, we follow the same strategy as proving Theorem 4.1. However, a significant challenge is introduced in that the linearization of F_C about m_{TW} is *not self-adjoint*, meaning that the classical techniques of Sect. 4 (based on self-adjointness of the linearization) no longer apply. Therefore, we develop a new method of proving linear stability based on the Grearhart-Prüss-Greiner (GPG) Theorem.

We decompose F_C as a sum of its linearization T_C about m_{TW} and its “nonlinear part” Ψ :

$$F_C(m_{TW} + u) = T_C u + \Psi(u). \quad (154)$$

The linearization about m_{TW} about traveling waves is:

$$T_C u := DF_C(m_{TW})u = u'' + \phi'(1/2)m'_{TW} + Vu' - (m_{TW}\phi')' - (u\phi'_{TW})', \quad (155)$$

where ϕ and ϕ_{TW} satisfy respectively

$$\begin{cases} -Z\phi'' + \phi = Pu & -1/2 < x < 1/2 \\ \phi(1/2) = \phi(-1/2) \\ \phi'(-1/2) = \phi'(1/2) \end{cases} \quad \text{and} \quad \begin{cases} -Z\phi''_{TW} + \phi_{TW} = Pm_{TW} & -1/2 < x < 1/2 \\ \phi_{TW}(1/2) = \phi_{TW}(-1/2) \\ \phi_{T,x}(-1/2) = \phi_{T,x}(1/2), \end{cases} \quad (156)$$

and $u \in \tilde{X}_C^2 := \left\{ m \in H^2(-1/2, 1/2) : m'(\pm 1/2) = 0, \int_{-1/2}^{1/2} u \, dx = 0 \right\}$. The coefficient V appears because the velocity of the traveling wave is $V = \phi'_{TW}(1/2)$.

Since the nonlinearity in F_C is quadratic (that is, the Keller-Segel term $(m\phi')'$), the nonlinear part about the traveling wave $m = m_{TW}$ is the same as the nonlinear part about the stationary state $m = 1$:

$$\Psi(u) = F_C(m_{TW} + u) - T_C u = \phi'(1/2)u' - (u\phi')', \quad (157)$$

with ϕ given in (156). Using the nonlinear part Ψ , and letting $u = m - m_{TW}$, we may rewrite the evolution equation (46) as

$$\partial_t u = T_C u + \Psi(u). \quad (158)$$

Similar to Sect. 4, our analysis in this section is focused on proving two key results:

1. 0 is an exponentially stable equilibrium of the linearized problem $u_t = T_C u$, and
2. Near m_{TW} , the linear part of F_C dominates the nonlinear part.

These results are given by Theorem 6.13 and Lemma 6.14, respectively. These two results are the traveling wave analogues of Theorem 4.3 and Lemma 4.9, and therefore once they are proved, the proof of Theorem 6.1 is identical to the proof of Theorem 4.1.

As described above however, a new challenge arises in this linearization: the operator T_C is non-self-adjoint, meaning that the spectral theorem used in the proof of linear stability in Theorem 4.3 no longer applies. Indeed, while, as we have already mentioned in the Introduction, a self-adjoint operator with compact inverse has a basis of eigenvectors, no such basis is guaranteed if the operator is non-self-adjoint operator, meaning that there may be a portion of the domain of the operator hidden from the eigenvectors. Since the action of the operator on this “dark” space cannot be determined from the eigenvectors, it is not sufficient merely to show that all the eigenvalues of the operator have negative real part. Instead, we rely on the GPG theorem [41], which we quote for convenience in Appendix C.

The GPG theorem overcomes the problem with invisibility of a part of the domain by considering not just eigenvalues, but the entire spectrum of the operator. The spectrum of a linear operator L is the set of all $\lambda \in \mathbb{C}$ so that the operator $\lambda I - L$ does not have a bounded inverse. Note that if $\lambda I - L$ is not invertible because it is not injective (one-to-one), then λ is an eigenvalue of L .

We recall that if L is a finite dimensional linear operator (a matrix), then the rank-nullity theorem applies and $\lambda I - L$ is invertible if and only if it is injective. In the infinite dimensional case, however, a linear operator may be injective but not surjective, and thus not invertible. Even if $\lambda I - L$ is invertible, its inverse may not be bounded. Thus, the spectrum of L may consist of more than just eigenvalues.

We also recall that if $\lambda I - L$ does have a bounded inverse, $(\lambda I - L)^{-1}$ is called a resolvent operator of L , and the set of λ such that the resolvent exists (that is the complement of the spectrum) is called the resolvent set. The solution $x(t)$ to the linear system $x_t = Lx$ can be written in terms of the resolvent via a line integral in the complex plane as an inverse Fourier Transform as

$$x(t) = \lim_{s \rightarrow \infty} \frac{1}{2\pi i} \int_{w-is}^{w+is} e^{\lambda t} (\lambda I - L)^{-1} x(0) d\lambda \quad (159)$$

for $w \in \mathbb{R}$ sufficiently large [41]. Furthermore, if $x(0)$ can be written $x(0) = \sum_{n=1}^{\infty} c_n x_n$ for eigenvectors x_n of L with eigenvalues λ_n , then if $w > \sup_n \operatorname{Re} \lambda_n$,

$$x(t) = \lim_{s \rightarrow \infty} \frac{1}{2\pi i} \int_{w-is}^{w+is} e^{\lambda t} (\lambda I - L)^{-1} \sum_{n=1}^{\infty} c_n x_n d\lambda \quad (160)$$

$$= \sum_{n=1}^{\infty} c_n x_n \lim_{s \rightarrow \infty} \frac{1}{2\pi i} \int_{w-is}^{w+is} \frac{e^{\lambda t}}{\lambda - \lambda_n} d\lambda \quad (161)$$

$$= \sum_{n=1}^{\infty} c_n x_n \frac{e^{wt}}{2\pi} \int_{-\infty}^{\infty} \frac{e^{ist}}{w + is - \lambda_n} ds \quad (162)$$

$$= \sum_{n=1}^{\infty} c_n x_n \frac{e^{wt}}{2\pi} \left(2\pi e^{(\lambda_n - w)t} \right) \quad (163)$$

$$= \sum_{n=1}^{\infty} c_n e^{\lambda_n t} x_n. \quad (164)$$

The crucial observation is that (159) holds even if $x(0)$ cannot be written as a sum of eigenvectors (i.e., if the eigenvectors of L do not span the domain of L) provided the resolvent $((w + is)I - L)^{-1}$ exists for all $s \in \mathbb{R}$ and w is sufficiently large.

The GRG theorem (formulated in Appendix C) provides conditions on the resolvent and spectrum of L such that, via (159), all solutions $x(t)$ converge to 0 exponentially fast. Then, since the solutions to our linearized problem $u_t = T_C u$ are $u(t) = S(t)u(0)$, we can conclude $\lim_{t \rightarrow \infty} \|u(t)\| \leq \lim_{t \rightarrow \infty} e^{-\sigma t} \|u(0)\| = 0$ whenever the conditions of the GRG theorem hold. Then 0 is exponentially stable in the linearized system.

In view of the above, to establish linear stability, it remains to be shown that each of the three conditions of the GPG theorem hold for the operator T_C . Those are identified in our Appendix C as conditions (i),(ii),(iii) and are also spelled out explicitly below. We will begin with proving condition (ii), then condition (i), and finally condition (iii). Before proceeding with these steps, we first show that T_C is non-self-adjoint.

Theorem 6.2. *There exists $Z^*, V^* > 0$ such that if $Z > Z^*$ and $0 < |V| < V^*$, the operator T_C is non-self-adjoint.*

Proof. As in the proof of Lemma 4.5, we will use the adjoint commutator. We will show that there exists $V^* > 0$ and $u_1, u_2 \in \tilde{X}_C^2$ so that if $0 < |V| < V^*$, then the adjoint commutator H for the operator $T_C = T_C(V)$ evaluated at u_1, u_2 is nonzero. This shows that $T_C(V)$ is non-self-adjoint.

Let $u_1(x) = \sin(\pi x)$ and $u_2(x) = \cos(2\pi x)$. Both u_1 and u_2 are in $\tilde{X}_C^2$. For $i = 1, 2$ let ψ_i satisfy $-Z\psi_i'' + \psi_i = P_{TW}(V)u_i$ with periodic boundary conditions in $(-1/2, 1/2)$. Then

$$\begin{aligned}
H(u_1, u_2) &= \int_{-1/2}^{1/2} u_1 u_2'' - u_2 u_1'' dx + \int_{-1/2}^{1/2} m'_{TW}(u_1 \psi_2'(1/2) \\
&\quad - u_2 \psi_1'(1/2)) dx + V \int_{-1/2}^{1/2} u_1 u_2' - u_2 u_1' dx \\
&\quad - \int_{-1/2}^{1/2} u_1 (m_{TW} \psi_2')' - u_2 (m_{TW} \psi_1')' dx \\
&\quad - \int_{-1/2}^{1/2} u_1 (u_2 \phi'_{TW})' - u_2 (u_1 \phi'_{TW})' dx.
\end{aligned} \tag{165}$$

The function ψ_1, ψ_2 depend on V through $P_{TW}(V)$, but from Lemma 5.2, $P'_{TW}(0) = 0$, so $\partial_V \psi_i|_{V=0} = 0$. Also from Proposition 5.2, the traveling wave solution satisfy $\partial_V m_{TW}|_{V=0} = m_1$ given by (102) and $\partial_V \phi_{TW}|_{V=0} = m_1(x) + x$. Therefore,

$$\begin{aligned}
\partial_V H(u_1, u_2)|_{V=0} &= \int_{-1/2}^{1/2} u_1 u_2'' - u_2 u_1'' dx + \int_{-1/2}^{1/2} m'_1(u_1 \psi_2'(1/2) \\
&\quad - u_2 \psi_1'(1/2)) dx + \int_{-1/2}^{1/2} u_1 u_2' - u_2 u_1' dx \\
&\quad - \int_{-1/2}^{1/2} u_1 (m_1 \psi_2')' - u_2 (m_1 \psi_1')' dx \\
&\quad - \int_{-1/2}^{1/2} u_1 (u_2 \phi'_1)' - u_2 (u_1 \phi'_1)' dx.
\end{aligned} \tag{166}$$

Since each of the functions $m_1, \phi_1, u_1, u_2, \psi_1$, and ψ_2 are explicitly known, and using $P_0 = \pi^2 Z + O(1)$ from Lemma 5.8, we may explicitly calculate the integrals in (166) and find the asymptotic expansion of the result in large Z :

$$\partial_V H(u_1, u_2)|_{V=0} = -3 + O(1/Z). \tag{167}$$

Therefore, for sufficiently large Z^* , if $Z > Z^*$, then $\partial_V H(u_1, u_2)|_{V=0} \neq 0$. Thus, there exists V^* so that if $0 < |V| < V^*$, $H(u_1, u_2) \neq 0$. We conclude that for $Z > Z^*$ and $0 < |V| < V^*$, $A_{TW}(V)$ is non-self-adjoint. $\square$

Condition (ii): the resolvent set of T_C contains the right half-plane, see appendix C. To establish that condition (ii) holds, we prove a sequence of four results. First, we show that the resolvent of T_C , if it exists, is compact. Then we show that for some $\lambda_0 > 0$, there exists a unique weak solution to $(\lambda_0 - T_C)u = w$ for each w , which implies that the resolvent $(\lambda_0 I - T_C)^{-1}$ exists. Next, we use the first two results to show that the spectrum of T_C consists only of its eigenvalues. Finally, we show that all the eigenvalues of T_C have negative real part. Thus, the resolvent set contains all complex numbers with positive real part, and condition (ii) is satisfied.

Proposition 6.3. *Suppose $\lambda \in \mathbb{C}$ such that $\lambda I - T_C$ is invertible. Then $(\lambda I - T_C)^{-1} : L^2(-1/2, 1/2) \rightarrow L^2(-1/2, 1/2)$ is a compact operator.*

Proof. To be compact, $(\lambda I - T_C)^{-1}$ must be bounded. Suppose, to the contrary, that it is unbounded. Then there exist sequences $(v_k) \subset \tilde{X}_C^2$ and $(w_k) \subset L^2(-1/2, 1/2)$ such that

$$(\lambda I - T_C)v_k = w_k, \quad \|v_k\|_{L^2} = 1, \quad \|w_k\|_{L^2} \leq 1/k. \quad (168)$$

Let ϕ_k satisfy $-Z\phi_k'' + \phi_k = Pv_k$ with periodic boundary conditions. Then the following sequence is bounded:

$$\begin{aligned} \langle w_k, v_k \rangle_{L^2} &= \langle (\lambda I - T_C)v_k, v_k \rangle_{L^2} \\ &= \lambda \|v_k\|_{L^2}^2 + \|v_k'\|_{L^2}^2 \\ &\quad + \langle (V - \phi_{TW}')v_k', v_k \rangle_{L^2} + \langle (\phi_k'(1/2) - \phi_k')m_{TW}', v_k \rangle_{L^2} \\ &\quad - \langle \phi_{TW}''v_k, v_k \rangle_{L^2} - \langle \phi_k''m_{TW}, v_k \rangle_{L^2}. \end{aligned} \quad (169)$$

Every term in this sequence is individually bounded due to Proposition 7.1 in Appendix A except possibly $\|v_k'\|_{L^2}^2$ and $\langle (V - \phi_{TW}')v_k', v_k \rangle_{L^2}$. However, the sum of these terms must be bounded. While the former is quadratic in $\|v_k'\|_{L^2}$, the latter is at most linear. Therefore, they must both be independently bounded as well.

Since $\|v_k\|_{L^2}$ and $\|v_k'\|_{L^2}$ are both bounded, we conclude that (v_k) is bounded with respect to the H^1 norm. The remaining arguments giving rise to a contradiction and proving that $(\lambda I - T_C)^{-1}$ is bounded and, moreover, compact, are identical to those in the proof of Lemma 4.7. $\square$

Proposition 6.4. *There exists $V^* > 0$ and $\lambda_0 \geq 0$ such that for each $w \in X^0$, there exists a unique weak solution $u \in X^1$ to $(\lambda_0 I - T_C)u = w$.*

Proof. Define the bilinear form $B : X^1 \times X^1 \rightarrow \mathbb{R}$ by

$$\begin{aligned} B[u, v] &= \langle u', v' \rangle_{L^2} - \langle (\phi'(1/2) - \phi')m_{TW}', v \rangle_{L^2} \\ &\quad + \langle (V - \phi_{TW}')u', m_{TW}\phi'' - u\phi_{TW}'', v \rangle_{L^2}. \end{aligned} \quad (170)$$

Then $u \in X^1$ is a weak solution to $(\lambda_0 I - T_C)u = w$ if and only if $\lambda_0 \langle u, v \rangle + B[u, v] = \langle w, v \rangle_{L^2}$ for all $v \in X^1$. We claim that there exist $a, b, V^* > 0$ and $\lambda_0 \geq 0$ such that if $|V| < V^*$, then

- $|B[u, v]| \leq a\|u\|_{H^1}\|v\|_{H^1}$
- $b\|v\|_{H^1}^2 \leq B[v, v] + \lambda_0\|v\|_{L^2}^2$.

The proof of these facts follows from the Poincaré inequality and the fact that $\|\phi_{TW}'\|_{L^2} = O(V)$. Therefore, by the Lax-Milgram Theorem, there exists a unique weak solution to $(\lambda_0 I - T_C)u = w$. $\square$

Proposition 6.5. *The spectrum of T_C consists only of its eigenvalues.*

Proof. This proof is essentially showing that the Fredholm alternative applies to T_C . Let $\lambda \in \mathbb{C}$, and let λ_0 be defined as in Proposition 6.4. Define $\hat{T}_C = \lambda_0 I - T_C$ and let $\lambda' = \lambda_0 - \lambda$. Then $\lambda I - T_C = \hat{T}_C - \lambda' I$. By Proposition 6.4, $\hat{T}_C$ is invertible, and by Proposition 6.3, $\hat{T}_C^{-1}$ is compact. Therefore, we may apply the Fredholm alternative for compact operators [36] to see that exactly one of the following holds:

- $(I - \lambda' \hat{T}_C^{-1})v = \hat{T}_C^{-1}w$ has a unique solution for each $w \in X^0$,
- $(I - \lambda' \hat{T}_C^{-1})v = 0$ has a nontrivial solution.

In either case, we may multiply by $\hat{T}_C$ to see that either $(\lambda I - T_C)v = w$ has a unique solution for all $w \in X^0$ or $(\lambda I - T_C)v = 0$. Therefore, either $\lambda I - T_C$ is invertible (with bounded inverse per Proposition 6.3) and therefore λ is not in the spectrum, or λ is an eigenvalue of T_C . Therefore, the spectrum of T_C consists only of its eigenvalues. $\square$

The following lemma shows that all eigenvalues of T_C have negative real part except possibly one. The following Theorem concerns this remaining eigenvalue showing that it too has negative real part, thus proving the desired result.

Lemma 6.6. *For V sufficiently small the eigenvalues of $T_C = T_C(V)$ all have negative real part bounded away from 0 except possibly one. Moreover, when $V = 0$, all the eigenvalues of $T_V(0)$ are negative (and real) except for a zero eigenvalue with multiplicity 1.*

Proof. The domain of $T_C(V)$ is $\tilde{X}_C^2$, which has the (Schauder) basis $\mathcal{B} = \{v_1, v_2, v_3, \dots\}$ where $v_n(x) = \sin(n\pi x)$ for n odd, and $v_n(x) = \cos(n\pi x)$ for n even. For each $n, m \in \mathbb{N}$, define

$$a_{mn} = \langle v_m, T_C(V)v_n \rangle_{L^2} \quad (171)$$

Treating $A = (a_{mn})$ as an “infinite matrix” operator on ℓ^2 , we see that λ is an eigenvalue of $T_C(V)$ if and only if $\lambda/2$ is an eigenvalue of A . In particular, the eigenvalues of A and $T_C(V)$ have the same sign.

Many of the terms in $T_C(V)$ vanish as $V \rightarrow 0$. In particular, the traveling waves m_{TW} and ϕ_{TW} and their derivatives depend smoothly on V in L^2 (see Remark 5.6). Moreover, when $V = 0$, m_{TW} and ϕ_{TW} are both constant in x (they are stationary states). Therefore, writing $m_{TW} = 1 + \tilde{m}_{TW}$, there exists $C_1, V^* > 0$ so that if $|V| < V^*$, then

$$\|m'_{TW}\|_{L^1}, \|\phi'_{TW}\|_{L^1}, \|\phi''_{TW}\|_{L^1}, \|\tilde{m}_{TW}\|_{L^1} \leq C_1|V|. \quad (172)$$

For each m , let ϕ_m solve $-Z\phi_m'' + \phi_m = P_{TW}(V)v_m$ with periodic boundary conditions in $(-1/2, 1/2)$. For each n, m , define

$$d_{mn} = \langle v_m, \phi_n'(1/2)m'_{TW} + Vv_n' - \tilde{m}_{TW}\phi_n'' - m'_{TW}\phi_n' - v_n'\phi_{TW}' - v_n\phi_{TW}'' \rangle_{L^2}. \quad (173)$$

From (172) and Lemma 6.8, there exists $C > 0$ independent of m such that

$$\sum_{n=1}^{\infty} |d_{mn}| \leq C|V| \quad \text{and} \quad \sum_{m=1}^{\infty} |d_{mn}| \leq C|V|. \quad (174)$$

Then we may write for each n, m :

$$a_{mn} = \langle v_m, T_C(0)v_n \rangle_{L^2} + d_{mn}. \quad (175)$$

The operator $T_C(0)$ (which is equal to $S_C(P_0)$) is defined by $T_C(0)u = u'' - \phi''$ where ϕ solves $-Z\phi'' + \phi = P_0u$ with periodic boundary conditions in $(-1/2, 1/2)$. Thus, letting $c_{mn} = \langle v_n, T_C(0)v_m \rangle_{L^2}$, we have $a_{mn} = c_{mn} + d_{mn}$. We can explicitly calculate c_{mn} :

$$c_{mn} = \begin{cases} -n^2\pi^2 + \frac{P_0}{Z} \frac{1}{1 + \frac{1}{\pi^2 n^2 Z}} & n = m \text{ even} \\ -n^2\pi^2 + \frac{P_0}{Z} \frac{1}{1 + \frac{1}{\pi^2 n^2 Z}} - \frac{4P_0 \coth\left(\frac{1}{2\sqrt{Z}}\right)}{\sqrt{Z}(1+n^2\pi^2 Z)^2} & n = m \text{ odd} \\ 0 & n \neq m \text{ either } m \text{ or } n \text{ even} \\ -\frac{4P_0(-1)^{\frac{m+1}{2} + \frac{n+1}{2}} \coth\left(\frac{1}{2\sqrt{Z}}\right)}{\sqrt{Z}(\pi^2 m^2 Z + 1)(\pi^2 n^2 Z + 1)} & n \neq m \text{ both odd.} \end{cases} \quad (176)$$

To show that all the eigenvalues of A are negative except possibly one of them we will use Theorem 3 of [94], which gives a Gershgorin-type result showing that all eigenvalues of an infinite matrix have negative real part. While possibly not all eigenvalues of A have negative real part, we will see using Theorem 3 of [94] that all eigenvalues of $D = B - I$ do have negative real part, and that all but one of these eigenvalues has real part less than -1 , thus proving the desired result.

The specific result of Theorem 3 of [94] is that there are countably many eigenvalues $\hat{\lambda}_n$ of $B = (b_{mn})$ and for each n ,

$$|\hat{\lambda}_n - b_{nn}| < Q_n := \sum_{\substack{m=1 \\ m \neq n}}^{\infty} |b_{mn}|, \quad (177)$$

provided the following conditions are met:

1. $b_{nn} \neq 0$ for any n and $\lim_{n \rightarrow \infty} |b_{nn}| = \infty$.
2. There exists $0 < \rho < 1$ so that for each odd n ,

$$\frac{Q_n}{|b_{nn}|} < \rho. \quad (178)$$

3. For each odd n, m with $n \neq m$, $|b_{nn} - b_{mm}| \geq Q_n + Q_m$
4. For each m , $\sup\{|b_{mn}| : n \in \mathbb{N}\} < \infty$.

We will show that B satisfies each of these conditions for small enough $V < V^*$ and large enough Z .

1. Observe that

$$b_{nn} < -n^2\pi^2 + \frac{P_0}{Z} \frac{1}{\frac{1}{n^2\pi^2 Z} + 1} + C|V| - 1. \quad (179)$$

Let $0 < \varepsilon < 1/(2 + 2\pi^2)$. Using Lemma 5.8, there exists Z^* large enough that for all $Z > Z^*$, $P_0/Z < \pi^2 + \varepsilon/2$. There also exists $V^* > 0$ so that if $|V| < V^*$, $C|V| < \varepsilon/2$. Therefore, for large enough Z , $b_{nn} < -\pi^2(n^2 - 1) - 1 + \varepsilon < 0$. It is clear that $\lim_{n \rightarrow \infty} |b_{nn}| = \infty$.

2. We have

$$Q_n = \sum_{\substack{m=1 \\ m \neq n}}^{\infty} |b_{mn}| \leq \sum_{\substack{m=1 \\ m \neq n}}^{\infty} |c_{mn}| + \sum_{\substack{m=1 \\ m \neq n}}^{\infty} |d_{mn}| \quad (180)$$

If n is even, $Q_n \leq \sum_{n=1}^{\infty} |d_{mn}| < C|V| < \varepsilon/2$. If n is odd, we can explicitly calculate a convenient upper bound for Q_n :

$$Q_n < Q_n + \frac{4P_0 \coth\left(\frac{1}{2\sqrt{Z}}\right)}{\sqrt{Z}(\pi^2 n^2 Z + 1)^2} \quad (181)$$

$$\leq \sum_{\substack{m=1 \\ m \text{ odd}}}^{\infty} \frac{4P_0 \coth\left(\frac{1}{2\sqrt{Z}}\right)}{\sqrt{Z}(\pi^2 m^2 Z + 1)(\pi^2 n^2 Z + 1)} + \sum_{m=1}^{\infty} |d_{mn}| \quad (182)$$

$$\leq \frac{P_0}{Z} \frac{1}{1 + n^2 \pi^2 Z} + C|V| \quad (183)$$

$$< \frac{\pi^2 + \varepsilon/2}{1 + \pi^2} + \frac{\varepsilon}{2} \quad \text{assuming } Z^* \geq 1. \quad (184)$$

We conclude that whether n is even or odd, (184) is an upper bound for Q_n . We have already seen that for $Z > Z^*$, $b_{nn} < -1 + \varepsilon$. Thus, for any n ,

$$\begin{aligned} \frac{Q_n}{|b_{nn}|} &< \frac{\pi^2 + \varepsilon/2}{(1 + \pi^2)(1 - \varepsilon)} + \frac{\varepsilon/2}{1 - \varepsilon} < \frac{\pi^2 + \frac{1}{4+4\pi^2}}{(1 + \pi^2)\left(1 - \frac{1}{2+2\pi^2}\right)} \\ &+ \frac{1}{(4 + 4\pi^2)\left(1 - \frac{1}{2+2\pi^2}\right)} = \frac{2 + 5\pi^2 + 4\pi^4}{2 + 6\pi^2 + 4\pi^4} < 1. \end{aligned} \quad (185)$$

Therefore, letting $\rho = \frac{2+5\pi^2+4\pi^4}{2+6\pi^2+4\pi^4}$, the second condition is satisfied.

3. Let $n, m \in \mathbb{N}$ be odd with $n \neq m$. One can verify that for any $Z > 0$,

$$0 < \frac{4\sqrt{Z} \coth\left(\frac{1}{2\sqrt{Z}}\right)}{(1 + \pi^2 Z)^2} < 1. \quad (186)$$

Then if $Z > Z^*$ and $Z > Z^*$,

$$|b_{nn} - b_{mm}| \geq \pi^2 |n^2 - m^2| - \frac{P_0}{Z} \left| \frac{1}{n^2 \pi^2 Z} + 1 - \frac{1}{m^2 \pi^2 Z} + 1 \right| \quad (187)$$

$$- \left| \frac{4P_0 \coth\left(\frac{1}{2\sqrt{Z}}\right)}{\sqrt{Z}(1 + n^2 \pi^2 Z)^2} - \frac{4P_0 \coth\left(\frac{1}{2\sqrt{Z}}\right)}{\sqrt{Z}(1 + m^2 \pi^2 Z)^2} \right| - 2C|V| \quad (188)$$

$$\geq \pi^2 |n^2 - m^2| - 2\frac{P_0}{Z} - \varepsilon \quad (189)$$

$$> 3\pi^2 - 2(\pi^2 - \varepsilon/2) - \varepsilon \quad (190)$$

$$> \pi^2 - 2\varepsilon. \quad (191)$$

On the other hand, we have seen that for each n , if $Z > Z^*$ and $|V| < V^*$, then $Q_n < 1$, so $Q_n + Q_m < 2 < \pi^2 - 2\varepsilon$. Thus condition 3 is satisfied.

4. This is clear.

Thus, the eigenvalues of B are enumerated $\hat{\lambda}_1, \hat{\lambda}_2, \hat{\lambda}_3, \dots$, and for each n , $|b_{nn} - \hat{\lambda}_n| < Q_n$. Thus, for $n \geq 2$,

$$\operatorname{Re} \hat{\lambda}_n < b_{nn} + Q_n < -4\pi^2 + \frac{P_0}{V_0} + C|V| < -3\pi^2 + \varepsilon < -1. \quad (192)$$

Since the eigenvalues of A are $\lambda_n = \hat{\lambda}_n + 1$, we conclude that all λ_n have negative real part bounded away from 0 except for possibly λ_1 . The eigenvalues of $T_C(V)$ are $2\lambda_n$ for $n = 1, 2, 3, \dots$, so the desired result holds.

In the case $V = 0$, the operator $T_C(0)$ is exactly the operator shown to have exactly one zero eigenvalue in the proof of Theorem 5.5. Therefore, $T_C(0)$ has all negative eigenvalues (real because the operator is self-adjoint) except for one zero eigenvalue. $\square$

Theorem 6.7. *There exists $V^*, Z^* > 0$ such that if $0 < |V| < V^*$ and $Z > Z^*$, then resolvent set of T_C contains $\{z \in \mathbb{C} : \operatorname{Re} z \geq 0\}$.*

Proof. Due to Proposition 6.5, we need only show that all eigenvalues of T_C have negative real part. Lemma 6.6 gives V^* and Z^* so that if $|V| < V^*$ and $Z > Z^*$, then all but possibly one of the eigenvalues of $T_C(V)$ has negative real part. We also know that when $V = 0$, this one eigenvalue is zero. Therefore, we only need to show that for $0 < |V| < V^*$, this eigenvalue has negative real part.

Since T_C depends on V , both explicitly, and through m_{TW} and ϕ_{TW} , we write $T_C = T_C(V)$. For the operator $T_C(V)$, the parameter $P = P_{TW}(V)$ is given by Theorem 5.5. We also consider the linearization $S_C(P)$ of F about $m = 1$ with arbitrary $P > 0$. We will make use of Corollary 1.13 and Theorem 1.16 in [26] which from which we conclude the following:

- There exists neighborhoods $U_1, U_2 \subset \mathbb{R}$ of 0 and $P_0 \in \mathbb{R}$ respectively and smooth functions $\lambda : U_1 \rightarrow \mathbb{R}$ and $\mu : U_2 \rightarrow \mathbb{R}$ such that $\lambda(V)$ is an eigenvalue of $T_C(V)$ and $\mu(P)$ is an eigenvalue of $T_C(P)$, and $\lambda(0) = \mu(P_0) = 0$.

- λ and μ satisfy:

$$-\mu'(P_0) \lim_{V \rightarrow 0} \frac{V P'_{TW}(V)}{\lambda(V)} = 1. \quad (193)$$

By Lemma 6.6, $\lambda(0) = 0$ is the largest eigenvalue of $T_C(0)$. Since T_C depends smoothly on V , so does $\lambda(V)$. Therefore, for small V , $\lambda(V)$ is the eigenvalue of $T_C(V)$ with the largest real part. Moreover, for small V , $\lambda(V)$ has the same sign as $-V P'(V) \mu'(P_0)$. From Proposition 5.8, $P'_{TW}(0) = 0$. For similar reasons, $\lambda'(0) = 0$. So after two applications of L'Hôpital's rule on (193), we obtain $\lambda(V) = \frac{1}{2} \lambda''(0) V^2 + O(V^3)$ and

$$\lambda''(0) = -2 P''_{TW}(0) \mu'(P_0). \quad (194)$$

Therefore, if $P''_{TW}(0) \mu'(P_0) > 0$, then there exists $V^* > 0$ such that if $0 < |V| < V^*$, then $\lambda(V) < 0$. We will show that for sufficiently large Z , both $P''_{TW}(0)$ and $\mu'(P_0)$ are positive, thus proving the desired result.

First we show that $\mu'(P_0)$ is positive. The eigenvalue equation satisfied by $\mu(P)$ is

$$u'' - \phi'' = \mu(P)u, \quad -Z\phi'' + \phi = Pu, \quad (195)$$

where $m(\pm 1/2) = 0$ and ϕ satisfies periodic boundary conditions. We write $P = P_0 + \varepsilon$ for some small ε , and expand m , ϕ , and μ in ε :

$$u = u_0 + \varepsilon u_1 + O(\varepsilon^2) \quad (196)$$

$$\phi = \phi_0 + \varepsilon \phi_1 + O(\varepsilon^2) \quad (197)$$

$$\mu = \mu_1 \varepsilon + O(\varepsilon^2). \quad (198)$$

Observe that $\mu_1 = \mu'(P_0)$. Solving the zeroth order in ε equation, we find u_0 and ϕ_0 up to a multiplicative constant:

$$u_0 = \frac{x}{P_0 - 1} - \frac{1}{2} \frac{P_0}{P_0 - 1} \csc\left(\frac{\sqrt{P_0 - 1}}{\sqrt{Z}}\right) \sin\left(\frac{\sqrt{P_0 - 1}x}{\sqrt{Z}}\right),$$

$$\phi_0 = u_0 + x. \quad (199)$$

Observe that, since P_0 and Z satisfy (96),

$$\csc\left(\frac{\sqrt{P_0 - 1}}{\sqrt{Z}}\right) = \sqrt{\frac{P_0^3 - P_0^2 + 4Z}{(P_0 - 1)P_0^2}}. \quad (200)$$

In first order, the (195) becomes

$$u_1'' - \phi_1'' = \mu_1 u_0, \quad -Z\phi_1'' + \phi_1 = P_0 u_1 + u_0. \quad (201)$$

Write $\phi_1 = \psi_1 + \phi_0/P_0$ where ψ_1 solves $-Z\psi_1'' + \psi_1 = P_0 u_1$. Thus, we may write the first order equation as

$$u_1'' - \psi_1'' = \mu_1 u_0 - \frac{1}{Z} u_0 + \frac{\phi_0}{P_0 Z}. \quad (202)$$

Since the operator $u_1 \mapsto u_1'' - \psi_1''$ (which is $S_C(P_0)$) is self adjoint, the right hand side must be orthogonal to the kernel of the operator, which is spanned by u_0 . Thus, μ_1 solves

$$\int_{-1/2}^{1/2} \left(\mu_1 u_0 - \frac{1}{Z} u_0 + \frac{\phi_0}{P_0 Z} \right) u_0 \, dx = 0. \quad (203)$$

Computing the integral and solving for μ_1 , we obtain

$$\mu_1 = \frac{3(P_0 - 1)(P_0^2 - 12Z)}{P_0 Z (3P_0^2 - 60Z + 2)}. \quad (204)$$

Using Lemma 5.8, we obtain an asymptotic form for μ_1 in large Z :

$$\mu_1 = \frac{1}{Z} + O(1/Z^2). \quad (205)$$

Thus, for sufficiently large Z , $\mu'(P_0) = \mu_1 > 0$.

Lemma 5.2 gives the value of P_2 . In large Z , this expands as

$$P_2 = \frac{\pi^2}{48} Z + O(1). \quad (206)$$

Thus, for large Z , $P_2 > 0$. Thus,

$$\lambda(V) = -\frac{\pi^2}{24} V^2 + O\left(\frac{V^4}{Z}\right), \quad (207)$$

so for large Z and small V , the largest real part of the eigenvalues of $T_C(V)$ is negative. $\square$

We conclude with a technical lemma used in the proof of Lemma 6.6.

Lemma 6.8. *Suppose $f : [-1/2, 1/2] \rightarrow \mathbb{R}$ is C^2 . Let $\mathcal{B} = \{v_1, v_2, v_3, \dots\}$ where $v_n(x) = \sin(n\pi x)$ for n odd, and $v_n(x) = \cos(n\pi x)$ for n even. Then there exists $C > 0$ such that*

$$\sum_{n=1}^{\infty} |\langle v_m, f v_n \rangle_{L^2}| < C \|f\|_{L^1} \quad \text{and} \quad \sum_{m=1}^{\infty} |\langle v_m, f v_n \rangle_{L^2}| < C \|f\|_{L^1}. \quad (208)$$

Proof. Decompose f as a Fourier series: $f = \sum_{k=1}^{\infty} a_k v_k$. Since f is C^2 -smooth, $|a_k| < \|f''\|_{L^1}/k^2$. Then we can use some product-to-sum trigonometric identities to see that

$$f v_n = \sum_{k=1}^{\infty} a_k v_k v_n = \sum_{k=1}^{\infty} \frac{a_k}{2} (r_{n,k} v_{k+n} + s_{n,k} v_{|k-n|}), \quad (209)$$

where the the coefficeints $r_{n,k}$ and $s_{n,k}$ are either 1 or -1 and are determined by the parities of n and k . The sign of each coefficient is not important, so we do not endeavor to give them explicitly. Thus,

$$\sum_{n=1}^{\infty} |\langle v_m, f v_n \rangle_{L^2}| = \sum_{n=1}^{\infty} \left| \sum_{k=1}^{\infty} \frac{a_k}{2} \langle v_m, r_{n,k} v_{k+n} + s_{n,k} v_{|k-n|} \rangle \right| \quad (210)$$

$$\leq \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \frac{|a_k|}{2} (|\langle v_m, v_{k+n} \rangle| + |\langle v_m, v_{|k-n|} \rangle|) \quad (211)$$

$$= \frac{1}{4} \sum_{n=1}^{\infty} |a_{n+m}| + |a_{|n-m|}| \quad (212)$$

$$\leq \frac{3}{4} \sum_{n=1}^{\infty} |a_n| \quad (213)$$

$$\leq \frac{3 \|f''\|_{L^1}}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} \quad (214)$$

$$= \frac{\pi^2}{8} \|f''\|_{L^1}. \quad (215)$$

Thus the result for the sum over n holds. The proof for the sum over m is identical. $\square$

Condition (i): T_C generates a strongly continuous semigroup, see Appendix C.

Here we show that the linearized operator T_C defined by (155) generates a strongly continuous semigroup. We will make use to of the Hille-Yosida Theorem [41]. We will first prove a supporting proposition.

Proposition 6.9. *There exists $V^*, Z^*, \lambda_0 > 0$ such that if $|V| < V^*$ and $Z > Z^*$, then all eigenvalues of for all $\lambda > 0$ and $u \in \tilde{X}_C^2$*

$$(\lambda - \lambda_0) \|u\|_{L^2} \leq \|(\lambda I - T_C)u\|_{L^2}. \quad (216)$$

Proof. We calculate the norm via the inner product:

$$\begin{aligned} \|(\lambda I - T_C)u\|_{L^2}^2 &= \langle (\lambda I - T_C)u, (\lambda I - T_C)u \rangle_{L^2} \\ &= \lambda^2 \|u\|_{L^2}^2 + \|T_C u\|_{L^2}^2 - 2\lambda \langle u, T_C u \rangle_{L^2}. \end{aligned}$$

Observe that

$$\begin{aligned} \langle u, T_C u \rangle_{L^2} &\leq -\|u'\|_{L^2}^2 + \|(\phi'(1/2) - \phi)m'_{TW}\|_{L^2}^2 \\ &\quad + \|(V - \phi_{TW})u'\|_{L^2}^2 + \|m_{TW}\phi''\|_{L^2}^2 + \|u\phi''_{TW}\|_{L^2}^2. \end{aligned} \quad (217)$$

There exists $C, V^* > 0$ so that $|V| < V^*$ so that (after applying the Poincaré inequality):

$$\begin{aligned} \|(\phi'(1/2) - \phi)m'_{TW}\|_{L^2}^2 &< C|V|\|u\|_{L^2}^2 \\ \|(V - \phi'_{TW})u'\|_{L^2}^2 &< C|V|\|u'\|_{L^2}^2 \\ \|m_{TW}\phi''\|_{L^2}^2 &\leq C\|u\|_{L^2}^2 \\ \|u\phi''_{TW}\|_{L^2}^2 &\leq C|V|\|u\|_{L^2}^2. \end{aligned} \quad (218)$$

Assume V^* is sufficiently small that $C|V| < 1/2$.

$$\langle u, T_C u \rangle_{L^2} \leq -\frac{1}{2}\|u'\|_{L^2}^2 + (1 + C)\|u\|_{L^2}^2. \quad (219)$$

Let $\lambda_0 = 2(1 + C)$ so that $\langle u, T_C u \rangle_{L^2} \leq (\lambda_0/2)\|u\|_{L^2}^2$. Thus,

$$\|(\lambda I - T_C)u\|_{L^2}^2 \geq \lambda^2\|u\|_{L^2}^2 - \lambda\lambda_0\|u\|_{L^2}^2 = (\lambda^2 - \lambda\lambda_0)\|u\|_{L^2}^2. \quad (220)$$

If $\lambda > \lambda_0$, then $\lambda^2 - \lambda\lambda_0 \geq \lambda^2 - 2\lambda\lambda_0 + \lambda_0^2 = (\lambda - \lambda_0)^2$. Therefore,

$$\|(\lambda I - T_C)u\|_{L^2} \geq (\lambda - \lambda_0)\|u\|_{L^2}. \quad (221)$$

□

We recall the definition of a closed operator.

Definition 6.10. Let X and Y be Banach spaces and let $B : D(B) \subset X \rightarrow Y$ be a linear operator. Then B is *closed* if for every sequence (x_n) converging to some $x \in X$ such that Bx_n converges to $y \in Y$, it follows that $x \in D(B)$ and $Bx = y$.

An operator is closed if its resolvent $(\lambda I - B)^{-1}$ exists and is bounded for at least one value of $\lambda \in \mathbb{C}$. By Theorem 6.7, the resolvent set of T_C is non-empty, and by Proposition 6.3, the resolvent is compact (and thus bounded) whenever it exists. Therefore, T_C is a closed operator. Thus, we may prove the main result of this section:

Proposition 6.11. *There exists $V^* > 0$ such that if $|V| < V^*$, then A generates a strongly continuous semigroup.*

Proof. We appeal the the Hille-Yosida Theorem [41], which states that if $T_C : X \rightarrow Y$ is a closed, densely defined operator and if there exists $\lambda_0 > 0$ such that

$$\|(\lambda I - T_C)^{-n}\|_{L^2} \leq \frac{1}{(\lambda - \lambda_0)^n}, \quad (222)$$

then T_C generates a strongly continuous semigroup.

It is clear to see that (222) is satisfied due to Proposition 6.9. Therefore, the hypotheses of the Hille-Yosida theorem are satisfied for sufficiently small V^* , so the result holds. □

Since T_C generates a strongly continuous semigroup, the first condition of the Gearhart-Prüss-Griener Theorem is satisfied.

Condition (iii): the resolvent of T_C is uniformly bounded, see appendix C.

Now we prove that the resolvent of T_C is uniformly bounded for complex numbers with positive real part. Then we formally establish linear stability in Theorem 6.13.

Proposition 6.12. *There exist $V^*, Z^*, \Gamma > 0$ such that if $0 < |V| < V^*$ and $Z > Z^*$, then the resolvent $\|(\lambda I - T_C)^{-1}\| < \Gamma$ for all $\lambda \in \mathbb{C}$ with $\operatorname{Re} \lambda > 0$.*

Proof. Existence of the resolvent $(\lambda I - T_C)^{-1}$ for all λ with $\operatorname{Re} \lambda > 0$ is established in Theorem 6.7. Assume, to the contrary, that there exists a sequence $(\lambda_k)_{k=1}^\infty \subset \mathbb{C}$ such that $\operatorname{Re} \lambda_k > 0$ for each k and

$$\|(\lambda_k I - T_C)^{-1}\|_{L^2} > k. \quad (223)$$

Then for each k , there exist $v_k \in \tilde{X}_C^2$ and $w_k \in L^2(-1/2, 1/2)$ such that $(\lambda_k I - T_C)v_k = w_k$, $\|v_k\|_{L^2} = 1$, and $\|w_k\|_{L^2} < 1/k$. We shall consider two cases: (i) the sequence (λ_k) is bounded, and (ii) (λ_k) is unbounded. We will show that in each case, we arrive at a contradiction.

- (i) If the sequence (λ_k) is bounded, then it has a subsequence also called (λ_k) which converges to some $\lambda \in \mathbb{C}$ with $\operatorname{Re} \lambda \geq 0$. By Theorem 6.7, λ is in the resolvent set of T_C . Recall the *first resolvent identity* [40] from which we conclude that for each k ,

$$(\lambda I - T_C)^{-1} - (\lambda_k I - T_C)^{-1} = (\lambda - \lambda_k)(\lambda I - T_C)^{-1}(\lambda_k I - T_C)^{-1}. \quad (224)$$

We calculate:

$$\begin{aligned} \|v_k\|_{L^2} &= \|(\lambda_k I - T_C)^{-1}w_k\|_{L^2} \\ &\leq \left\| - \left[(\lambda I - T_C)^{-1} - (\lambda_k I - T_C)^{-1} \right] w_k \right\|_{L^2} + \|(\lambda I - T_C)^{-1}w_k\|_{L^2} \\ &\leq \|(\lambda_k - \lambda)(\lambda I - T_C)^{-1}(\lambda_k I - T_C)^{-1}w_k\|_{L^2} + \|w_k\|_{L^2}\|(\lambda I - T_C)^{-1}\| \\ &\leq |\lambda_k - \lambda| \left\| (\lambda I - T_C)^{-1}v_k \right\|_{L^2} + \|w_k\|_{L^2}\|(\lambda I - T_C)^{-1}\| \\ &\leq (|\lambda_k - \lambda| \|v_k\|_{L^2} + \|w_k\|_{L^2}) \|(\lambda I - T_C)^{-1}\|_{L^2}. \end{aligned}$$

Since $|\lambda_k - \lambda_0|, \|w_k\|_{L^2} \rightarrow 0$ and $\|v_k\|_{L^2}$ is bounded, we conclude that $\|v_k\|_{L^2} \rightarrow 0$, a contradiction. Therefore, (λ_k) is not bounded.

- (ii) If the sequence (λ_k) is unbounded, then it has a subsequence also called (λ_k) such that $\lambda_k \rightarrow \infty$. There exists corresponding sequences (v_k) and (w_k) such that

$$w_k = (\lambda_k I - T_C)v_k, \quad \|v_k\|_{L^2} = 1, \quad \|w_k\|_{L^2} \leq 1/k. \quad (225)$$

We calculate the inner product

$$\begin{aligned} \langle w_k, v_k \rangle_{L^2} &= \lambda_k + \|v'_k\|_{L^2} + \int_{-1/2}^{1/2} (V - \phi'_{TW}) v'_k \bar{v}_k \, dx \\ &\quad + \int_{-1/2}^{1/2} m'_{TW} (\phi'_k(1/2) - \phi'_k) \bar{v}_k \, dx - \int_{-1/2}^{1/2} \phi''_{TW} |v_k|^2 \, dx \\ &\quad - \int_{-1/2}^{1/2} m_{TW} \phi''_k \bar{v}_k \, dx \end{aligned} \quad (226)$$

Since (v_k) is L^2 -bounded, by Proposition 7.1 in Appendix A, the last three integrals in (226) are uniformly bounded:

$$\left| \int_{-1/2}^{1/2} m'_{TW} (\phi'_k(1/2) - \phi'_k) \bar{v}_k \, dx - \int_{-1/2}^{1/2} \phi''_{TW} |v_k|^2 \, dx - \int_{-1/2}^{1/2} m_{TW} \phi''_k \bar{v}_k \, dx \right| < C \quad (227)$$

for some $C > 0$ independent of k .

Taking the real part of (226), we find using the Cauchy-Schwartz inequality and the Poincaré inequality that

$$\operatorname{Re} \langle w_k, v_k \rangle \geq \operatorname{Re} \lambda_k + \|v'_k\|_{L^2} - \frac{1}{\pi} \|V - \phi'_{TW}\|_{L^\infty} \|v'_k\|_{L^2}^2 - C. \quad (228)$$

Assuming V^* is sufficiently small that if $|V| < V^*$, then $\|V - \phi'_{TW}\|_{L^\infty} < \pi$, we conclude that $\operatorname{Re} \langle w_k, v_k \rangle \geq \operatorname{Re} \lambda_k - C$. On the other hand, $\operatorname{Re} \langle w_k, v_k \rangle \leq |\langle w_k, v_k \rangle| < 1/k$. Since $\operatorname{Re} \lambda_k > 0$, we conclude that $(\operatorname{Re} \lambda_k)$ is bounded. Furthermore, since all terms in (228) have been shown to be bounded except those involving $\|v'_k\|$, we conclude that (v'_k) must be bounded as well.

Now taking the imaginary part of (226), we find that

$$\operatorname{Im} \langle w_k, v_k \rangle \geq \operatorname{Im} \lambda_k - \frac{1}{\pi} \|V - \phi'_{TW}\|_{L^\infty} \|v'_k\|_{L^2}^2 - C. \quad (229)$$

Once again, all terms in this equation are known to be bounded in k except $\operatorname{Im} \lambda_k$, so we conclude that $(\operatorname{Im} \lambda_k)$ is bounded also, a contradiction.

Since (λ_k) can be neither bounded nor unbounded, we conclude that no such sequence (λ_k) can exist, and so $(\lambda I - T_C)^{-1}$ is uniformly bounded. That is, there exists $\Gamma > 0$ such that

$$\|(\lambda I - T_C)^{-1}\| < \Gamma. \quad (230)$$

□

Now that we have in place all the results proving the conditions of the GPG theorem, we may apply it to prove linear stability.

Theorem 6.13. *There exist $V^*, Z^*, \Gamma, \sigma > 0$ such that if $|V| < V^*$ and $Z > Z^*$ then T_C generates a strongly continuous semigroup $\{S(t) : t \geq 0\}$ satisfying*

$$\|S(t)\| < \Gamma e^{-\sigma t}. \quad (231)$$

Proof. We need to satisfy the three hypotheses of the GPG theorem 9.1, see Appendix C. We have checked that indeed:

- (i) is satisfied for sufficiently small V^* due to Proposition 6.11
- (ii) is satisfied for sufficiently small V^* and sufficiently large Z^* due to Theorem 6.7.
- (iii) is satisfied due to Proposition 6.12.

Thus, the desired result holds. $\square$

The last key piece of the proof of Theorem 6.1 is that the linear part of F_C dominates the nonlinear part near m_{TW} , which is proved by the following lemma:

Lemma 6.14. *Let $T, \delta > 0$ and let u be a solution to*

$$\begin{cases} \partial_t u = T_C u + \Psi(u) & -1/2 < x < 1/2, \ 0 < t < T \\ u' = 0 & x = \pm 1/2, \ t > 0. \end{cases} \quad (232)$$

There exist $V^, U^* > 0$ such that if $\|u'(0, \cdot)\|_{L^2} < \delta$, $|V| < V^*$, and $\|u(\cdot, t)\|_{L^2} < U^*$ for all $0 \leq t \leq T$, then*

$$\|u'(\cdot, t)\|_{L^2} \leq \delta \quad (233)$$

for all $0 \leq t \leq T$.

Proof. Write the evolution equation (46) as

$$\partial_t u - u'' = Bu + \Psi(m). \quad (234)$$

Where B is defined by

$$\begin{aligned} Bu &= (\phi'(1/2) - \phi')m'_{TW} + (V - \phi'_{TW})u' - m_{TW}\phi'' \\ -u\phi''_{TW}, \quad &\begin{cases} -Z\phi'' + \phi = Pu \\ \phi(-1/2) = \phi(1/2) \\ \phi'(-1/2) = \phi'(1/2). \end{cases} \end{aligned} \quad (235)$$

Now square both sides and integrate to obtain

$$\|Bu + \Psi(u)\|_{L^2}^2 = \int_{-1/2}^{1/2} \Psi^2(u) \, dx \quad (236)$$

$$= \int_{-1/2}^{1/2} (\partial_t u)^2 - 2(\partial_t u)u'' + (u'')^2 \, dx \quad (237)$$

$$= \|\partial_t u\|_{L^2}^2 + 2 \int_{-1/2}^{1/2} (\partial_t u')u' \, dx + \|m''\|_{L^2}^2 \quad (238)$$

$$= \|\partial_t u\|_{L^2}^2 + \frac{d}{dt} \|u'\|_{L^2}^2 + \|u''\|_{L^2}^2. \quad (239)$$

Thus,

$$\frac{d}{dt} \|u'\|_{L^2}^2 \leq \|Bu + \Psi(u)\|_{L^2}^2 - \|u''\|_{L^2}^2 \quad (240)$$

$$\leq 2\|Bu\|_{L^2}^2 + 2\|\Psi(u)\|_{L^2}^2 - \|u''\|_{L^2}^2. \quad (241)$$

From Lemma 4.8, there exists C_1 independent of u , V , and Z such that

$$\|\Psi(u)\|_{L^2} \leq C_1 \|u\|_{L^2} \|u\|_{H^1}. \quad (242)$$

Observe that due to 7.1, if $|V| < V^*$ is small enough, there exist C_2, C_3 depending only on Z such that

$$\|Bu\|_{L^2} \leq C_2 V^* \|u'\|_{L^2} + C_3 \|u\|_{L^2}. \quad (243)$$

Since $\int_{-1/2}^{1/2} u \, dx = 0$ and $u'(\pm 1/2, t) = 0$, we may apply the Poincaré inequality to both u and u' with a Poincaré constant of π :

$$\pi \|u\|_{L^2} \leq \|u'\|_{L^2} \quad \text{and} \quad \pi \|u'\|_{L^2} \leq \|u''\|_{L^2}. \quad (244)$$

Thus,

$$\frac{d}{dt} \|u'\|_{L^2}^2 \leq 2C_1^2 \|u\|_{L^2}^2 \|u\|_{H^1}^2 + 4C_2^2 (V^*)^2 \|u'\|_{L^2}^2 + 4C_3^2 \|u\|_{L^2}^2 - \|u''\|_{L^2}^2 \quad (245)$$

$$\leq 2C_1^2 \left(1 + \frac{1}{\pi}\right)^2 \|u\|_{L^2}^2 \|u'\|_{L^2}^2 + 4C_2^2 (V^*)^2 \|u'\|_{L^2}^2 + 4C_3^2 \|u\|_{L^2}^2 - \|u''\|_{L^2}^2 \quad (246)$$

$$\leq -\left(\pi^2 - 4C_1^2 \|u\|_{L^2}^2 - 4C_2^2 (V^*)^2\right) \|u'\|_{L^2}^2 + 4C_3^2 \|u\|_{L^2}^2. \quad (247)$$

Without loss of generality, we may assume that

$$V^* \leq \frac{\pi}{4C_2} \quad \text{and} \quad U^* \leq \frac{\pi}{4C_1}. \quad (248)$$

Then, if $\|u\|_{L^2} \leq U^*$ for all $0 < t < T$,

$$\frac{d}{dt} \|u'\|_{L^2}^2 \leq -R_1 \|u'\|_{L^2}^2 + R_2 \quad (249)$$

where

$$R_1 = \pi^2 - 4C_1^2 (U^*)^2 - 4C_2^2 (V^*)^2 \geq \frac{\pi^2}{2} \quad \text{and} \quad R_2 = 4C_3^2 (U^*)^2. \quad (250)$$

We now introduce a new variable:

$$q(t) = \|u'(\cdot, t)\|_{L^2}^2 - \frac{R_2}{R_1}. \quad (251)$$

Then q satisfies $q' \leq -R_1 q$. By Grönwall's inequality,

$$q(t) \leq q(0)e^{-R_1 t}. \quad (252)$$

We conclude that if $q(0) < 0$, then $q(t) < 0$ for all $t > 0$. Thus, if $|V| < V^*$ and $\|u\|_{L^2} \leq U^*$, and if

$$\|u'(\cdot, 0)\|_{L^2} < \sqrt{\frac{R_2}{R_1}}, \quad \text{then} \quad \|u'(\cdot, t)\|_{L^2} < \sqrt{\frac{R_2}{R_1}} \quad (253)$$

for all $t > 0$. Letting U^* be sufficiently small that

$$\sqrt{\frac{R_2}{R_1}} \leq \frac{\sqrt{8}}{\pi} C_3 U^* < \delta, \quad (254)$$

the desired result holds. $\square$

With Lemma 6.14 in place, we may duplicate the proof of Theorem 4.1 in order to prove the nonlinear stability of traveling waves via Theorem 6.1.

Acknowledgements. We thank V. Rybalko for many helpful discussions on NSA and the relevance in this case of the GPG theorem. We also thank O. Krupchytskyi for his feedback on the proofs and mathematical techniques used in this paper. Finally, we thank J.-F. Joanny, J. Casademunt and P. Recho for discussing the physical aspects of the model and the subtlety of stability in the problems with NSA.

Funding L. B. was supported by the National Science Foundation grants DMS-2005262 and DMS-2404546. A. S. was also partially supported by the same National Science Foundation grant DMS-2005262. L.T. acknowledges the support under the French grants ANR-17-CE08-0047-02, ANR-21-CE08-MESOCRYSP and the European grant ERC-H2020-MSCA-RISE-2020-101008140.

Data Availability This manuscript does not report any datasets or generated data. All results are theoretical, and no data were used or produced in this study.

Declarations

Conflict of interest The authors declare that they have no Conflict of interest or Conflict of interest associated with this work.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

7. Appendix A

Here we show the Proposition which controls the solution ϕ to (34), (37) and (38).

Proposition 7.1. *Let $u \in L^0([-1/2, 1/2])$. Then there exists a unique solution $\phi \in W^{2,p}(-1/2, 1/2)$ for any $1 \leq p \leq \infty$ satisfying $-Z\phi'' + \phi = Pu$ with periodic boundary conditions on $(-1/2, 1/2)$. Moreover, ϕ satisfies the following for any $1 \leq p \leq \infty$:*

- $\|\phi\|_{L^p} \leq P\|u\|_{L^p}$,
- $\|\phi'\|_{L^\infty} \leq \frac{P}{2\sqrt{Z}}\|u\|_{L^2}$,
- $\|\phi''\|_{L^p} \leq \frac{2P}{Z}\|u\|_{L^p}$.

Proof. The solution ϕ can be calculated explicitly using a Green's function:

$$\begin{aligned}\phi(x) &= \frac{P}{2\sqrt{Z} \sinh\left(\frac{1}{2\sqrt{Z}}\right)} \int_{-1/2}^{1/2} G(x, y) u(y) dy, \\ G(x, y) &= \begin{cases} \cosh\left(\frac{1/2+(y-x)}{\sqrt{Z}}\right) & y < x \\ \cosh\left(\frac{1/2+(x-y)}{\sqrt{Z}}\right) & y > x. \end{cases}\end{aligned}\quad (255)$$

By Young's Integral inequality [85], $\|\phi\|_{L^p} \leq C\|u\|_{L^p}$ where

$$\begin{aligned}C &= \sup_{|x| \leq 1/2} \frac{P}{2\sqrt{Z} \sinh\left(\frac{1}{2\sqrt{Z}}\right)} \int_{-1/2}^{1/2} |G(x, y)| dy \\ &= \sup_{|y| \leq 1/2} \frac{P}{2\sqrt{Z} \sinh\left(\frac{1}{2\sqrt{Z}}\right)} \int_{-1/2}^{1/2} |G(x, y)| dx.\end{aligned}\quad (256)$$

We calculate

$$\frac{P}{2\sqrt{Z} \sinh\left(\frac{1}{2\sqrt{Z}}\right)} \int_{-1/2}^{1/2} |G(x, y)| dy \quad (257)$$

$$\begin{aligned}&= \frac{P}{2\sqrt{Z} \sinh\left(\frac{1}{2\sqrt{Z}}\right)} \left(\int_{-1/2}^x \cosh\left(\frac{1/2+y-x}{\sqrt{Z}}\right) dy \right. \\ &\quad \left. + \int_x^{1/2} \cosh\left(\frac{1/2+x-y}{\sqrt{Z}}\right) dy \right)\end{aligned}\quad (258)$$

$$\begin{aligned}&= \frac{P}{2 \sinh\left(\frac{1}{2\sqrt{Z}}\right)} \left(\sinh\left(\frac{1}{2\sqrt{Z}}\right) + \sinh\left(\frac{x}{\sqrt{Z}}\right) \right. \\ &\quad \left. - \sinh\left(\frac{x}{\sqrt{Z}}\right) + \sinh\left(\frac{1}{2\sqrt{Z}}\right) \right)\end{aligned}\quad (259)$$

$$= P. \quad (260)$$

We conclude that $\|\phi\|_{L^p} \leq P\|u\|_{L^p}$. Next, since G is continuous and differentiable in x except where $x = y$,

$$\phi'(x) = \frac{P}{2\sqrt{Z} \sinh\left(\frac{1}{2\sqrt{Z}}\right)} \int_{-1/2}^{1/2} \frac{d}{dx} G(x, y) u(y) dy. \quad (261)$$

Therefore, using Hölder's inequality for p and its Hölder conjugate q ,

$$\|\phi'\|_{L^\infty} \leq \frac{P}{2\sqrt{Z} \sinh\left(\frac{1}{2\sqrt{Z}}\right)} \|u\|_{L^p} \sup_{|x| \leq 1/2} \left\| \frac{d}{dx} G(x, \cdot) \right\|_{L^q}. \quad (262)$$

Since $|\frac{d}{dx} G(x, y)| \leq \frac{1}{\sqrt{Z}} \sinh\left(\frac{1}{2\sqrt{Z}}\right)$, we conclude that $\|\phi'\|_{L^\infty} \leq \frac{P}{2Z} \|u\|_{L^p}$. Finally, since $\phi'' = -\frac{P}{Z}u + \frac{1}{Z}\phi$, we have

$$\|\phi''\|_{L^p} \leq \frac{P}{Z} \|u\|_{L^p} + \frac{1}{Z} \|\phi\|_{L^p} \leq \frac{2P}{Z} \|u\|_{L^p}. \quad (263)$$

□

8. Appendix B

Here we formulate for convenience the Crandall-Rabinowitz (CR) theorem [25].

Theorem 8.1. *Let X and Y be Banach spaces, and let $\mathcal{F} : X \times \mathbb{R} \rightarrow Y$ be an operator with the following properties:*

- $\mathcal{F}(0, t) = 0$ for all t .
- $D_x \mathcal{F}$, $D_t \mathcal{F}$, and $D_{xt} \mathcal{F}$ exist and are continuous.
- The dimension of the null space and co-dimension of the range of $D_x \mathcal{F}(0, 0)$ are both 1.
- If $x_0 \neq 0$ is in the null space of $D_x \mathcal{F}(0, 0)$, then $D_{xt} \mathcal{F}(0, 0)x_0$ is not in the range of $D_x \mathcal{F}(0, 0)$.

Then there exists a neighborhood $U \subset X \times \mathbb{R}$ of $(0, 0)$, $\varepsilon > 0$ and functions $\sigma : (-\varepsilon, \varepsilon) \rightarrow X$ and $s : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}$ with $\sigma \not\equiv 0$ such that $\sigma(0) = 0$, $s(0) = 0$, and

$$\mathcal{F}^{-1}(0) \cap U = (\{(0, t) : t \in \mathbb{R}\} \cup \{(\sigma(\alpha), s(\alpha)) : |\alpha| < \varepsilon\}) \cap U. \quad (264)$$

Moreover, if $\mathcal{F}_{xx}$ exists and is continuous, then σ is continuously differentiable and $\sigma'(0)$ spans the null space of $D_x \mathcal{F}(0, 0)$.

In the main text we proceed by checking systematically these four properties for the operator (134).

9. Appendix C

Here we formulate the the Gearhart-Prüss-Greiner (GPG) theorem [41].

Theorem 9.1. *Let X be a Hilbert space, and let $L : D(L) \rightarrow X$ be a linear operator, where the domain $D(L)$ of L is a dense subspace of X . If the following conditions hold*

- (i) *the semigroup $(S(t))_{t \geq 0}$ generated by L is strongly continuous,*
- (ii) *The resolvent set of L contains $\{z \in \mathbb{C} : \operatorname{Re} z > 0\}$, and*
- (iii) *The resolvent $(\lambda I - L)^{-1}$ is uniformly bounded on the above set, i.e.,*

$$\sup_{\operatorname{Re} \lambda > 0} \|(\lambda I - L)^{-1}\|_X < \infty, \quad (265)$$

then there exists $\Gamma, \sigma > 0$ such that

- (a) *For each λ in the spectrum $\sigma(S(t))$, $|\lambda| < e^{-\sigma t}$, and*
- (b) *For each $t \geq 0$, $\|S(t)\|_X \leq \Gamma e^{-\sigma t}$.*

In the main text we proceed by checking systematically these three conditions for the operator (155).

References

1. AGARWAL, P., ZAIDEL-BAR, R.: Diverse roles of non-muscle myosin II contractility in 3D cell migration. *Essays Biochem.* **63**(5), 497–508, 2019
2. ALBERTS, B., JOHNSON, A., LEWIS, J., RAFF, M., ROBERTS, K., WALTER, P.: Molecular biology of the cell. *OCLC* **145080076**(48122761), 57023651, 2002
3. ALERT, R., BLANCH-MERCADER, C., CASADEMUNT, J.: Active fingering instability in tissue spreading. *Phys. Rev. Lett.* **122**(8), 088104, 2019
4. ARROWSMITH, D., PLACE, C.M.: Dynamical Systems: Differential Equations, Maps, and Chaotic Behaviour, vol. 5. CRC Press, Boca Raton (1992)
5. ASHIDA, Y., GONG, Z., UEDA, M.: Non-hermitian physics. *Adv. Phys.* **69**(3), 249–435, 2020
6. AULBACH, B., WANNER, T.: The Hartman-Grobman theorem for Carathéodory-type differential equations in Banach spaces. *Nonlinear Anal. Theory Methods Appl.* **40**(1–8), 91–104, 2000
7. BARKER, B., HUMPHERYS, J., LYNG, G., LYTLE, J.: Evans function computation for the stability of travelling waves. *Philos. Trans. R. Soc. A Math. Phys. Eng. Sci.* **376**(2117), 20170184, 2018
8. BARNHART, E., LEE, K.-C., ALLEN, G.M., THERIOT, J.A., MOGILNER, A.: Balance between cell- substrate adhesion and myosin contraction determines the frequency of motility initiation in fish keratocytes. *Proc. Natl. Acad. Sci.* **112**(16), 5045–5050, 2015
9. BARNHART, E.L., ALLEN, G.M., JÜLICHER, F., THERIOT, J.A.: Bipedal locomotion in crawling cells. *Biophys. J.* **98**(6), 933–942, 2010
10. BARNHART, E.L., LEE, K.-C., KEREN, K., MOGILNER, A., THERIOT, J.A.: An adhesion-dependent switch between mechanisms that determine motile cell shape. *PLoS Biol.* **9**(5), e1001059, 2011
11. BENDOUKHA, S., ABDELMALEK, S., KIRANE, M.: The global existence and asymptotic stability of solutions for a reaction-diffusion system. *Nonlinear Anal. Real World Appl.* **53**, 103052, 2020
12. BERLYAND, L., FUHRMANN, J., RYBALKO, V.: Bifurcation of traveling waves in a keller-segel type free boundary model of cell motility. *Commun. Math. Sci.* **16**(3), 735–762, 2018
13. BERLYAND, L., POTOMKIN, M., RYBALKO, V.: Phase-field model of cell motility: traveling waves and sharp interface limit. *C.R. Math.* **354**(10), 986–992, 2016
14. BERLYAND, L., POTOMKIN, M., RYBALKO, V.: Sharp interface limit in a phase field model of cell motility (2017)
15. BERNACKI, M.: Vertex and front-tracking methods for the modeling of microstructure evolution at the solid state. *Digital Materials: Continuum Numerical Methods at the Mesoscopic Scale*, pp. 155–176 (2024)
16. BOIS, J.S., JÜLICHER, F., GRILL, S.W.: Pattern formation in active fluids. *Biophys. J.* **100**(3), 445a, 2011
17. BOURBAKI, N.: General Topology: Chapters 5–10, vol. 18. Springer Science & Business Media, Berlin (2013)
18. BRAY, D.: Cell Movements: From Molecules to Motility. Garland Science, New York City (2000)
19. CALLAN-JONES, A.C., VOITURIEZ, R.: Active gel model of amoeboid cell motility. *New J. Phys.* **15**(2), 025022, 2013
20. CALVEZ, V., HAWKINS, R.J., MEUNIER, N., VOITURIEZ, R.: Analysis of a nonlocal model for spontaneous cell polarization. *SIAM J. Appl. Math.* **72**(2), 594–622, 2012
21. CARLSSON, A.E., SEPT, D.: Mathematical modeling of cell migration. *Methods Cell Biol.* **84**, 911–937, 2008
22. CHELLY, H., RECHO, P.: Cell motility as an energy minimization process. *Phys. Rev. E* **105**, 064401, 2022

23. PAUL CORNWELL AND CHRISTOPHER KRT JONES: A stability index for travelling waves in activator-inhibitor systems. *Proc. R. Soc. Edinburgh Sect. A Math.* **150**(1), 517–548, 2020
24. COWAN, J.M., DUGGAN, J.J., HEWITT, B.R., PETRIE, R.J.: Non-muscle myosin ii and the plasticity of 3d cell migration. *Front. Cell Dev. Biol.* **10**, 1047256, 2022
25. CRANDALL, M.G., RABINOWITZ, P.H.: Bifurcation from simple eigenvalues. *J. Funct. Anal.* **8**(2), 321–340, 1971
26. CRANDALL, M.G., RABINOWITZ, P.H.: Bifurcation, perturbation of simple eigenvalues and linearized stability (1973)
27. CRANDALL, M.G., RABINOWITZ, P.H.: Mathematical theory of bifurcation. In: Bifurcation Phenomena in Mathematical Physics and Related Topics: Proceedings of the NATO Advanced Study Institute held at Cargèse, Corsica, pp. 3–46. Springer (1980)
28. CUCCHI, A., MELLET, A., MEUNIER, N.: A cahn-hilliard model for cell motility. *SIAM J. Math. Anal.* **52**(4), 3843–3880, 2020
29. CUCCHI, A., MELLET, A., MEUNIER, N.: Self polarization and traveling wave in a model for cell crawling migration. *Discrete Dyn. Syst.* **42**, 2381–2407, 2022
30. DAVID, N., PERTHAME, B.: Free boundary limit of a tumor growth model with nutrient. *Journal de Mathématiques Pures et Appliquées* **155**, 62–82, 2021
31. DE BELLY, H., WEINER, O.D.: Follow the flow: actin and membrane act as an integrated system to globally coordinate cell shape and movement. *Curr. Opin. Cell Biol.* **89**, 102392, 2024
32. DE SOUSA, F.S., MANGIAVACCHI, N., NONATO, L.G., CASTELO, A., TOMÉ, M.F., FERREIRA, V.G., CUMINATO, J.A., MCKEE, S.: A front-tracking/front-capturing method for the simulation of 3d multi-fluid flows with free surfaces. *J. Comput. Phys.* **198**(2), 469–499, 2004
33. DIMILLA, P.A., BARBEE, K., LAUFFENBURGER, D.A.: Mathematical model for the effects of adhesion and mechanics on cell migration speed. *Biophys. J.* **60**(1), 15–37, 1991
34. DINELLI, A., O'BYRNE, J., CURATOLO, A., ZHAO, Y., SOLLICH, P., TAILLEUR, J.: Non-reciprocity across scales in active mixtures. *Nat. Commun.* **14**(1), 7035, 2023
35. DOLPH, C.L.: Recent developments in some non-self-adjoint problems of mathematical physics. *Bull. Am. Math. Soc.* **67**(1), 1–69, 1961
36. DOUGLAS, R.G.: Banach Algebra Techniques in Operator Theory, vol. 179. Springer Science & Business Media, Berlin (2012)
37. DRIVER, B.K.: Analysis Tools with Applications. Springer, New York (2003)
38. XINXIN, D., DOUBROVINSKI, K., OSTERFIELD, M.: Self-organized cell motility from motor-filament interactions. *Biophys. J.* **102**(8), 1738–1745, 2012
39. DUAN, YU., AGUDO-CANALEJO, J., GOLESTANIAN, R., MAHAULT, B.: Dynamical pattern formation without self-attraction in quorum-sensing active matter: the interplay between nonreciprocity and motility. *Phys. Rev. Lett.* **131**(14), 148301, 2023
40. DUNFORD, N.: Linear Operators, Part 1: General Theory, vol. 10. John Wiley & Sons, Hoboken (1988)
41. ENGEL, K.-J., NAGEL, R., BRENDLE, S.: One-Parameter Semigroups for Linear Evolution Equations, vol. 194. Springer, Berlin (2000)
42. EVANS, J.W.: Nerve axon equations: III stability of the nerve impulse. *Indiana Univ. Math. J.* **22**(6), 577–593, 1972
43. FARUTIN, A., ÉTIENNE, J., MISBAH, C., RECHO, P.: Crawling in a fluid. *Phys. Rev. Lett.* **123**, 118101, 2019
44. FORTUNATO, I.C., SUNYER, R.: The forces behind directed cell migration. *Biophysica* **2**(4), 548–563, 2022
45. FRIEDMAN, A., BEI, H.: Asymptotic stability for a free boundary problem arising in a tumor model. *J. Differ. Equ.* **227**(2), 598–639, 2006
46. FRIEDMAN, A., BEI, H.: Bifurcation for a free boundary problem modeling tumor growth by stokes equation. *SIAM J. Math. Anal.* **39**(1), 174–194, 2007

47. GIOMI, L., DESIMONE, A.: Spontaneous division and motility in active nematic droplets. *Phys. Rev. Lett.* **112**(14), 147802, 2014
48. GUO, S., WU, J.: Bifurcation Theory of Functional Differential Equations, vol. 10. Springer, Berlin (2013)
49. HAO, W., HAUSTEIN, J.D., BEI, H., LIU, Y., SOMMESE, A.J., ZHANG, Y.-T.: Bifurcation for a free boundary problem modeling the growth of a tumor with a necrotic core. *Nonlinear Anal. Real World Appl.* **13**(2), 694–709, 2012
50. HAWKINS, R.J., POINCELOUX, R., BÉNICHOU, O., PIEL, M., CHAVRIER, P., VOITURIEZ, R.: Spontaneous contractility-mediated cortical flow generates cell migration in three-dimensional environments. *Biophys. J.* **101**(5), 1041–1045, 2011
51. HOWARD, J.: Mechanics of motor proteins. In: Physics of bio-molecules and cells. Physique des biomolécules et des cellules: session LXXV, pp. 69–94. Springer (2002)
52. NIKOLAI IVANOVICH IONKIN: The stability of a problem in the theory of heat conduction with nonclassical boundary conditions. *Differ. Uravn.* **15**(7), 1279–1283, 1979
53. JUELICHER, F., KRUSE, K., PROST, J., JOANNY, J.-F.: Active behavior of the cytoskeleton. *Phys. Rep.* **449**(1–3), 3–28, 2007
54. KAPITULA, T.: On the stability of traveling waves in weighted L^∞ spaces. *J. Differ. Equ.* **112**(1), 179–215, 1994
55. KAPITULA, T., PROMISLOW, K.; et al.: Spectral and Dynamical Stability of Nonlinear Waves, vol. 457. Springer, Berlin (2013)
56. KELLER, E.F., SEGEL, L.A.: Model for chemotaxis. *J. Theor. Biol.* **30**(2), 225–234, 1971
57. KEREN, K., PINCUS, Z., ALLEN, G.M., BARNHART, E.L., MARRIOTT, G., MOGILNER, A., THERIOT, J.A.: Mechanism of shape determination in motile cells. *Nature* **453**(7194), 475–480, 2008
58. KIMPTON, L.S., WHITELEY, J.P., WATERS, S.L., KING, J.R., OLIVER, J.M.: Multiple travelling-wave solutions in a minimal model for cell motility. *Math. Med. Biol. J. IMA* **30**(3), 241–272, 2013
59. KOWALL, C., MARCINIAK-CZOCRA, A., MÜNNICH, F.: Stability results for bounded stationary solutions of reaction-diffusion-ode systems. arXiv preprint [arXiv:2201.12748](https://arxiv.org/abs/2201.12748) (2022)
60. KREYSZIG, E.: Introductory Functional Analysis with Applications, vol. 17. John Wiley & Sons, Hoboken (1991)
61. KRUSE, K., JOANNY, J.-F., JÜLICHER, F., PROST, J.: Contractility and retrograde flow in lamellipodium motion. *Phys. Biol.* **3**(2), 130, 2006
62. KUMAR, R., HIREMATH, K.R., MANZETTI, S.: A primer on eigenvalue problems of non-self-adjoint operators. *Anal. Math. Phys.* **14**(2), 21, 2024
63. LARRIPA, K., MOGILNER, A.: Transport of a 1d viscoelastic actin-myosin strip of gel as a model of a crawling cell. *Phys. A* **372**(1), 113–123, 2006
64. LATUSHKIN, Y., SUKHTAYEV, A.: The algebraic multiplicity of eigenvalues and the Evans function revisited. *Math. Model. Nat. Phenomena* **5**(4), 269–292, 2010
65. LEDOUX, V., MALHAM, S.J.A., NIESEN, J., THÜMLER, V.: Computing stability of multidimensional traveling waves. *SIAM J. Appl. Dyn. Syst.* **8**(1), 480–507, 2009
66. LOOSLEY, A.J., TANG, J.X.: Stick-slip motion and elastic coupling in crawling cells. *Phys. Rev. E-Stat. Nonlinear Soft Matter Phys.* **86**(3), 031908, 2012
67. KENING, L.: A Hartman-Grobman theorem for scalar reaction-diffusion equations. *J. Differ. Equ.* **93**(2), 364–394, 1991
68. MARÉE, A.F.M., JILKINE, A., DAWES, A., GRIENEISEN, V.A., EDELSTEIN-KESHET, L.: Polarization and movement of keratocytes: a multiscale modelling approach. *Bull. Math. Biol.* **68**, 1169–1211, 2006
69. MEISS, J.D.: Differential Dynamical Systems. SIAM, Philadelphia (2007)
70. MOGILNER, A.: Mathematics of cell motility: have we got its number? *J. Math. Biol.* **58**, 105–134, 2009

71. NICKAEEN, M., NOVAK, I.L., PULFORD, S., RUMACK, A., BRANDON, J., SLEPCHENKO, B.M., MOGILNER, A.: A free-boundary model of a motile cell explains turning behavior. *PLoS Comput. Biol.* **13**(11), e1005862, 2017
72. NWOGBAGA, I., CAMLEY, B.A.: Coupling cell shape and velocity leads to oscillation and circling in keratocyte galvanotaxis. *Biophys. J.* **122**(1), 130–142, 2023
73. OHTA, T., TARAMA, M., SANO, M.: Simple model of cell crawling. *Phys. D* **318**, 3–11, 2016
74. PAYNE, L.E., WEINBERGER, H.F.: An optimal poincaré inequality for convex domains. *Arch. Ration. Mech. Anal.* **5**(1), 286–292, 1960
75. PEGO, R.L., WEINSTEIN, M.I.: Eigenvalues, and instabilities of solitary waves. *Philos. Trans. R. Soc. Lond. Ser. A: Phys. Eng. Sci.* **340**(1656), 47–94, 1992
76. PELINOVSKY, D.E.: Inertia law for spectral stability of solitary waves in coupled nonlinear Schrödinger equations. *Proc. R. Soc. A Math. Phys. Eng. Sci.* **461**(2055), 783–812, 2005
77. PUTELAT, T., RECHO, P., TRUSKINOVSKY, L.: Mechanical stress as a regulator of cell motility. *Phys. Rev. E* **97**, 012410, 2018
78. RAFELSKI, S.M., THERIOT, J.A.: Crawling toward a unified model of cell motility: spatial and temporal regulation of actin dynamics. *Annu. Rev. Biochem.* **73**(1), 209–239, 2004
79. RECHO, P., PUTELAT, T., TRUSKINOVSKY, L.: Contraction-driven cell motility. *Phys. Rev. Lett.* **111**(10), 108102, 2013
80. RECHO, P., PUTELAT, T., TRUSKINOVSKY, L.: Mechanics of motility initiation and motility arrest in crawling cells. *J. Mech. Phys. Solids* **84**, 469–505, 2015
81. RECHO, P., TRUSKINOVSKY, L.: Asymmetry between pushing and pulling for crawling cells. *Phys. Rev. E-Stat. Nonlinear Soft Matter Phys.* **87**(2), 022720, 2013
82. RUBINSTEIN, B., FOURNIER, M.F., JACOBSON, K., VERKHOVSKY, A.B., MOGILNER, A.: Actin-myosin viscoelastic flow in the keratocyte lamellipod. *Biophys. J.* **97**(7), 1853–1863, 2009
83. RUBINSTEIN, B., JACOBSON, K., MOGILNER, A.: Multiscale two-dimensional modeling of a motile simple-shaped cell. *Multiscale Model. Simul.* **3**(2), 413–439, 2005
84. RUDIN, W.: Functional Analysis. McGraw-Hill, Columbus, New York (1991)
85. RUSSO, B.: On the Hausdorff-Young theorem for integral operators (1977)
86. RYBALKO, V., BERLYAND, L.: Emergence of traveling waves and their stability in a free boundary model of cell motility. *Trans. Am. Math. Soc.* **376**(03), 1799–1844, 2023
87. ALEX SAFSTEN, C., RYBALKO, V., BERLYAND, L.: Asymptotic stability of contraction-driven cell motion. *Phys. Rev. E* **105**(2), 024403, 2022
88. SANDSTEDE, B.: Chapter 18 - stability of travelling waves. In: Bernold, F. (Ed) Handbook of Dynamical Systems, volume 2 of Handbook of Dynamical Systems, pp. 983–1055. Elsevier Science (2002)
89. SCHREIBER, C.H., STEWART, M., DUKE, T.: Simulation of cell motility that reproduces the force-velocity relationship. *Proc. Natl. Acad. Sci.* **107**(20), 9141–9146, 2010
90. SCHWARTZ, C.A., YAN, A.: Construction of Lyapunov functions for nonlinear systems using normal forms. *J. Math. Anal. Appl.* **216**(2), 521–535, 1997
91. SENGUPTA, S., PARENT, C.A., BEAR, J.E.: The principles of directed cell migration. *Nat. Rev. Mol. Cell Biol.* **22**(8), 529–547, 2021
92. SHAAT, M., PARK, H.S.: Chiral nonreciprocal elasticity and mechanical activity. *J. Mech. Phys. Solids* **171**, 105163, 2023
93. SHELLARD, A., MAYOR, R.: All roads lead to directional cell migration. *Trends Cell Biol.* **30**(11), 852–868, 2020
94. SHIVAKUMAR, P.N., WILLIAMS, J.J., RUDRAIAH, N.: Eigenvalues for infinite matrices. *Linear Algebra Appl.* **96**, 35–63, 1987
95. TJHUNG, E., TIRIBOCCHI, A., MARENDUZZO, D., CATES, M.E.: A minimal physical model captures the shapes of crawling cells. *Nat. Commun.* **6**(1), 5420, 2015
96. TREFETHEN, L.N.: Spectra and pseudospectra: the behavior of nonnormal matrices and operators (2020)

97. VICENTE-MANZANARES, M., MA, X., ADELSTEIN, R.S., HORWITZ, A.R.: Non-muscle myosin ii takes centre stage in cell adhesion and migration. *Nat. Rev. Mol. Cell Biol.* **10**(11), 778–790, 2009
98. WANG, Q., YANG, X., ADALSTEINSSON, D., ELSTON, T.C., JACOBSON, K., KAPUSTINA, M., FOREST, M.G.: Computational and modeling strategies for cell motility. *Computational Modeling of Biological Systems: From Molecules to Pathways*, pp. 257–296 (2012)
99. WEISSENBRUCH, K., MAYOR, R.: Actomyosin forces in cell migration: Moving beyond cell body retraction. *Bioessays*, pp. 2400055 (2024)
100. WOLGEMUTH, C.W., STAJIC, J., MOGILNER, A.: Redundant mechanisms for stable cell locomotion revealed by minimal models. *Biophys. J.* **101**(3), 545–553, 2011
101. QIAN, W., XIANCHEN, X., QIAN, H., WANG, S., ZHU, R., YAN, Z., MA, H., CHEN, Y., HUANG, G.: Active metamaterials for realizing odd mass density. *Proc. Natl. Acad. Sci.* **120**(21), e2209829120, 2023
102. YANG, Y., JOLLY, M.K., LEVINE, H.: Computational modeling of collective cell migration: mechanical and biochemical aspects. *Cell Migr. Causes Funct.*, 2019. https://doi.org/10.1007/978-3-030-17593-1_1
103. ZIEBERT, F., ARANSON, I.S.: Effects of adhesion dynamics and substrate compliance on the shape and motility of crawling cells. *PLoS ONE* **8**(5), e64511, 2013
104. ZIEBERT, F., ARANSON, I.S.: Computational approaches to substrate-based cell motility. *npj Comput. Mater.* **2**(1), 1–16, 2016

Leonid Berlyand

Department of Mathematics and Huck Institute for Life Sciences,
The Pennsylvania State University,
University Park
USA.

and

Alex Safsten

Department of Mathematics,
University of Maryland,
College Park
USA.

e-mail: safsten@umd.edu

and

Leoni Berlyand ESPCI Paris,
Paris
France.

(Received November 17, 2024 / Accepted November 14, 2025)

Published online December 10, 2025

© The Author(s) (2025)